Riemannian Diffusion Matérn Kernels: Learned Geometry from a Unimodular Metric Field

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Abstract

Gaussian process kernels encode geometry, but in most nonstationary constructions it is unclear which geometric object is actually identifiable from the learned covariance. We study operator-defined Matérn priors driven by a learned Riemannian diffusion and introduce the *Riemannian Diffusion Matérn* (RDM) kernel family. In RDM-Geo, a unimodular metric field $g(x)$ enters the SPDE operator $L_{\kappa,g} = \kappa^2 - \Delta_g$, defining the covariance operator $\mathcal{K} = \sigma^2 L_{\kappa,g}^{-\alpha}$ and learning geometry directly in the input domain without a latent embedding. Our main result is kernel-level identifiability: the one-parameter kernel family $\{k(\cdot,\cdot;\kappa)\}_{\kappa>0}$ is an injective Laplace transform of the heat kernel and thus uniquely determines the semigroup, generator, and (under the unimodular gauge) the diffusion geometry. Varadhan asymptotics further link short-range correlation decay to intrinsic geodesic distance. We demonstrate geometry learning on synthetic domains and on Great Lakes surface temperature data using sparse SPDE–FEM inference.

1. Introduction

Nonstationary Gaussian process (GP) models are widely used when correlation structure is shaped by geometry: lake temperatures respect coastlines, turbulence depends on altitude and prevailing directions, and stresses propagate along material anisotropy. In such settings, *the kernel is the geometry*: correlation should decay with an intrinsic distance, potentially anisotropic and spatially varying. This has motivated a large literature on nonstationary kernels (Sampson & Guttorp, 1992; Gibbs, 1997; Higdon, 1998; Paciorek & Schervish, 2003; 2006; Fuglstad et al., 2015a; Wilson et al., 2016; de Souza et al., 2023; Saad et al., 2024), including diffusion-kernel constructions (Kondor & Lafferty, 2002; Lafferty & Lebanon, 2005).

Most existing work takes a *forward* view: propose a valid nonstationary covariance rule and train it for prediction. What is typically missing is the *inverse* question: from a learned GP law, *which geometric object is uniquely determined*? This matters for interpretability and for understanding intrinsic non-identifiabilities known in Matérn-type estimation (Zhang, 2004; Fuglstad et al., 2015b). Two dominant modelling strategies highlight the tension. Warping methods (including DKL and deep GPs) learn an embedding map (Wilson et al., 2016; de Souza et al., 2023), but without strong constraints it can fold the domain and collapse distant regions (Ober et al., 2021). Operator-first approaches instead define the field directly on the input domain using (nonstationary) SPDEs (Lindgren et al., 2011; Fuglstad et al., 2015a; Lindgren et al., 2022), with sparse discretisations and scalable inference (Lindgren et al., 2011; Fuglstad et al., 2015a; Bolin & Kirchner, 2020; Bolin et al., 2024); however, the corresponding geometry learning problem remains ill-posed unless the intrinsic degrees of freedom are made explicit.

Geometric Matérn priors based on the Laplace–Beltrami operator make the geometry explicit: the diffusion tensor acts as an (inverse) metric and correlation decay is driven by intrinsic distance. Yet most Matérn-on-manifold results take geometry as fixed (Borovitskiy et al., 2020; Li et al., 2023) or learn it indirectly in a latent space (Jørgensen & Hauberg, 2021; Niu et al., 2023; Fichera et al., 2023). Even when learning a metric field on the input domain, identifiability is subtle: microergodicity issues arise already for classical Matérn estimation (Zhang, 2004), our operator parameterisation has an exact scale degeneracy (Proposition 4.3), and density/volume effects can change amplitudes without changing leading exponential decay (Varadhan, 1967). Our approach is to make the inverse map itself explicit: by treating κ as a resolvent parameter, the *kernel family* uniquely determines the heat kernel and generator via Laplace-transform inversion (Widder, 1941), yielding a rigorous kernel-level identifiability criterion for the unimodular diffusion geometry (Section 3).

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We instantiate this programme with the *Riemannian Diffusion Matérn* (RDM) family, defined through spectral calculus for elliptic operators (Davies, 1989) and a pointwise unimodular gauge that separates intrinsic anisotropy from determinant/scale effects. We learn a smooth unimodular SPD field using an INR in global log coordinates (Arsigny et al., 2006; Pennec et al., 2006; Tancik et al., 2020; Sitzmann et al., 2020) and train by marginal likelihood via sparse SPDE–FEM inference with Hutchinson trace estimators and stochastic Lanczos quadrature (SLQ) log-determinants (Hutchinson, 1990; Ubaru et al., 2017). We evaluate on synthetic geometry-recovery tasks and on Great Lakes surface temperatures over complex shorelines. A detailed review of related work is provided in Appendix A.

1) We frame nonstationary kernel learning as an inverse problem and define which diffusion geometry is learnable from the GP law (Section 3).
2) We introduce RDM-Geo with a pointwise unimodular gauge, resolving scale degeneracies to isolate the intrinsic geometry governing short-range correlation decay (Assumption 4.4 and Theorem 3.12).
3) We prove a kernel-level identifiability theorem: the kernel family is an injective transform of the heat kernel and uniquely determines the generator and unimodular geometry (Theorem 3.2 and Corollary 3.3); we validate the resulting geometry learning empirically.

2. Preliminaries

2.1. Notation and analytic setup

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. We use standard notation for $L^2(\Omega)$ inner products and Sobolev spaces $H^s(\Omega)$. Define $\mathrm{SPD}^1(d) := \{A \in \mathrm{SPD}(d) : \det A = 1\}$. For an inverse metric field $A = g^{-1}$, we denote the induced geodesic distance by d_g and assume homogeneous Neumann boundary conditions and weak PDE formulations throughout; definitions extend to the Dirichlet case (H_0^1) naturally.

Assumption 2.1 (Uniform ellipticity and boundedness)**.** The (inverse) metric field $A : \Omega \to \mathbb{R}^{d\times d}$ is symmetric and there exist constants $0 < a \le b < \infty$ such that $aI \preceq A(x) \preceq bI$ for almost every $x \in \Omega$. Let $\kappa > 0$ be a fixed scalar.

2.2. Gaussian processes, kernels, and covariance operators

Given a continuous symmetric positive-definite kernel k, the associated covariance operator $\mathcal{K} : L^2(\Omega) \to L^2(\Omega)$ is

$$(\mathcal{K}g)(x) = \int_\Omega k(x, x')\, g(x')\, \mathrm{d}x'. \tag{1}$$

If $\mathcal{K}$ is self-adjoint, positive, and trace-class, it admits an integral kernel and eigen-expansion with nonnegative spectrum (Mercer when k is continuous). We work in this regime for pointwise kernels (for RDM, $\alpha > \frac{d}{2}$); for general $\alpha > 0$ the operator construction still defines a valid covariance operator and hence a (possibly distribution-valued) Gaussian field (see Appendix B). For RDM-Full, we work on the weighted space $L^2(\Omega, \rho\, \mathrm{d}x)$, with kernels understood with respect to $\rho\, \mathrm{d}x$.

2.3. Elliptic operators from bilinear forms

Define the bilinear form $a : H^1(\Omega) \times H^1(\Omega) \to \mathbb{R}$ by

$$a(u, v) := \kappa^2(u, v) + \int_\Omega (A(x)\nabla u) \cdot \nabla v\, \mathrm{d}x, \tag{2}$$

with the domain modified to $H_0^1(\Omega)$ under Dirichlet conditions. Under Assumption 2.1, a is bounded and coercive. We define $L_{A,\kappa}$ as the (unbounded) operator on $L^2(\Omega)$ associated with a via the Friedrichs extension: $u \in \mathcal{D}(L_{A,\kappa})$ iff there exists $f \in L^2(\Omega)$ such that $a(u, v) = (f, v)$ for all $v \in H^1(\Omega)$, and we set $L_{A,\kappa}u = f$. Then $L_{A,\kappa}$ is self-adjoint and strictly positive on $L^2(\Omega)$ and satisfies

$$(L_{A,\kappa}u, v) = a(u, v) \qquad \forall u \in \mathcal{D}(L_{A,\kappa}),\ v \in H^1(\Omega). \tag{3}$$

On bounded domains, the resolvent is compact and the spectrum is discrete (see, e.g., Gilbarg & Trudinger 2001).

For reference, the differential expression associated with (3) is the variable-coefficient elliptic operator

$$L_{A,\kappa} = \kappa^2 - \nabla\cdot\big(A(x)\nabla\big), \tag{4}$$

with domain depending on the boundary condition (Neumann or Dirichlet).

Geometry and Laplace–Beltrami. When $A(x)$ is interpreted as the inverse of a Riemannian metric $g(x)$, i.e. $A(x) = g(x)^{-1}$, the Laplace–Beltrami operator on (Ω, g) is

$$\Delta_g f \;=\; \frac{1}{\sqrt{|g|}}\, \partial_i\Big(\sqrt{|g|}\, g^{ij}\, \partial_j f\Big), \qquad |g| = \det g. \tag{5}$$

If g is at least C^1, then $g^{ij} = A^{ij}$ and the product rule gives

$$\Delta_g f = \nabla\cdot(A\nabla f) + \langle A\nabla \log\sqrt{|g|}, \nabla f\rangle,$$

i.e. the difference between Δ_g and the divergence-form operator is an explicit first-order drift induced by the volume element. Under the unimodular gauge $|g| \equiv 1$ (equivalently $\det A \equiv 1$), the drift term vanishes and $\Delta_g f = \nabla\cdot(A\nabla f)$. Thus the RDM operator $L_{\kappa,g} := \kappa^2 - \Delta_g$ coincides with $L_{A,\kappa}$ in (4) under Assumptions 2.1 and 4.4.

2.4. Spectral theorem, Weyl asymptotics, and trace-class criteria

On bounded domains, $L_{A,\kappa}$ is self-adjoint and strictly positive with discrete spectrum $\{(\lambda_j, \varphi_j)\}_{j\geq 1}$, and fractional powers are defined by spectral calculus. Weyl asymptotics imply $\lambda_j \asymp j^{\frac{2}{d}}$ (e.g., Davies 1989); hence $L_{A,\kappa}^{-\alpha}$ is trace-class iff $\alpha > \frac{d}{2}$.

Proposition 2.2 (Trace-class threshold)**.** *Let $L_{A,\kappa}$ satisfy Assumption 2.1. Then for any $\alpha > 0$, $L_{A,\kappa}^{-\alpha}$ is positive and self-adjoint on $L^2(\Omega)$ and is trace-class iff $\alpha > \frac{d}{2}$.*

2.5. Fractional powers and the heat semigroup

For $\alpha > 0$, the fractional powers of $L_{A,\kappa}$ are defined via spectral calculus as above and also through the Balakrishnan formula

$$L_{A,\kappa}^{-\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty t^{\alpha-1} e^{-tL_{A,\kappa}} \, \mathrm{d}t, \tag{6}$$

where $\{e^{-tL_{A,\kappa}}\}_{t\geq 0}$ is the analytic, positivity-preserving heat semigroup with integral kernel $p_t(x, x')$ (see, e.g., Pazy 1983). The kernel is symmetric $p_t(x, x') = p_t(x', x)$; it is jointly continuous for $t > 0$ and becomes smooth under additional coefficient regularity. Under Neumann boundary conditions one has $\int_\Omega p_t(x, x') \, \mathrm{d}x' = e^{-\kappa^2 t}$, reflecting the "mass decay" induced by the κ^2 term.

2.6. Matérn covariances and the constant-coefficient SPDE

Let $A(x) \equiv I$ and set $L_0 = \kappa^2 - \Delta$. Consider the SPDE

$$L_0^{\frac{\alpha}{2}} f = \mathcal{W}, \qquad \alpha = \nu + \tfrac{d}{2}. \tag{7}$$

Solving in the eigenbasis of L_0 gives $\langle f, \varphi_j \rangle = \lambda_j^{-\frac{\alpha}{2}} \xi_j$ and thus

$$\mathcal{K} = L_0^{-\alpha}, \qquad k(x, x') = \sum_{j\geq 1} \lambda_j^{-\alpha} \varphi_j(x)\varphi_j(x'). \tag{8}$$

In $\mathbb{R}^d$ this recovers the classical Matérn form with smoothness ν; on bounded domains, it is the restricted-domain analogue up to boundary effects (Lindgren et al., 2011).

3. Main Results

Taking the inverse view, we formalise what geometry is *identifiable* from the GP law (equivalently, from the kernel family) and characterise the intrinsic geometry encoded by the generator.

Definition 3.1 (Learnable diffusion geometry)**.** In RDM-Geo we learn a unimodular inverse metric field $A(x) = g(x)^{-1} \in \mathrm{SPD}^1(d)$, which defines the diffusion geometry $g = A^{-1}$. In RDM-Full we learn a general SPD field $\tilde{A}(x) \in \mathrm{SPD}(d)$ and define its determinant split pointwise by

$$\rho(x) := \big(\det \tilde{A}(x)\big)^{1/d}, \qquad \bar{A}(x) := \rho(x)^{-1}\tilde{A}(x),$$

so that $\bar{A}(x) \in \mathrm{SPD}^1(d)$ and $\tilde{A} = \rho\,\bar{A}$ (formalised in Proposition 3.6). The *intrinsic* diffusion geometry is the unimodular metric $\bar{g} = \bar{A}^{-1}$. Varadhan asymptotics show that the correlation geometry (exponential decay distance) depends only on $\bar{g}$ and is invariant to pointwise conformal rescalings of $\tilde{A}$ (Theorem 3.12 and Proposition 3.11).

We now state the main kernel-level identifiability result for RDM-Geo (Lebesgue reference measure); the RDM-Full decomposition clarifies what part of the geometry is intrinsic, but its absolute density scale is confounded with σ.

Kernel families (RDM-Geo). Fix $\alpha > \frac{d}{2}$ so that RDM-Geo admits a pointwise Mercer kernel. For each $\kappa > 0$, let $k(\cdot, \cdot; \kappa)$ denote the integral kernel of the covariance operator $\mathcal{K}_\kappa = \sigma^2(\kappa^2 - \Delta_g)^{-\alpha}$ (equivalently $\mathcal{K}_\kappa = \sigma^2(\kappa^2 - \nabla\cdot(A\nabla))^{-\alpha}$ under the unimodular gauge). The collection $\{k(\cdot, \cdot; \kappa)\}_{\kappa>0}$ is the *kernel family*. Treating κ as a resolvent parameter is what makes the inverse map sharp: a single κ admits exact scale confounding (Proposition 4.3), whereas the full family is injective.

Theorem 3.2 (Kernel family $\Longleftrightarrow$ heat kernel $\Longleftrightarrow$ generator (RDM-Geo))**.** *Assume Ω has smooth boundary and the unimodular diffusion tensor $A = g^{-1}$ is smooth and uniformly elliptic so that the corresponding heat kernel is continuous and satisfies standard Gaussian bounds. Let $\mathcal{L} := \Delta_g$ (equivalently $\mathcal{L} = \nabla\cdot(A\nabla)$ under the unimodular gauge), with reflecting (no-flux) boundary conditions. Let $q_t(x, y)$ denote the heat kernel of $e^{t\mathcal{L}}$ with respect to Lebesgue measure, i.e.*

$$(e^{t\mathcal{L}} f)(x) = \int_\Omega q_t(x, y)\, f(y) \, \mathrm{d}y.$$

Then for every $\kappa > 0$ and $x, y \in \Omega$,

$$k(x, y; \kappa) = \frac{\sigma^2}{\Gamma(\alpha)} \int_0^\infty t^{\alpha-1} e^{-\kappa^2 t} \, q_t(x, y) \, \mathrm{d}t. \tag{9}$$

Moreover, the map $q_\bullet \mapsto \{k(\cdot, \cdot; \kappa)\}_{\kappa>0}$ is injective: if two RDM-Geo models have identical kernel families, $k_1(x, y; \kappa) = k_2(x, y; \kappa)$ for all $\kappa > 0$ and all $x, y \in \Omega$, then $q_t^{(1)}(x, y) = q_t^{(2)}(x, y)$ for all $t > 0$ and all $x, y \in \Omega$, and consequently the semigroups coincide and the generators are equal: $\mathcal{L}_1 = \mathcal{L}_2$.

Corollary 3.3 (Identifiability of the diffusion tensor and geometry (RDM-Geo))**.** *Under the assumptions of Theorem 3.2, the kernel family uniquely identifies the diffusion tensor in the generator. Under the unimodular*

gauge $\det g \equiv 1$*, the generator is* $\mathcal{L} = \nabla \cdot (A\nabla)$ *with* $A = g^{-1} \in \mathrm{SPD}^1(d)$*. If two RDM-Geo models have identical kernel families, then* $A_1(x) = A_2(x)$ *a.e. on* Ω *(hence* $g_1 = g_2$ *a.e.).*

Remark 3.4 (Identifiability versus finite-data estimation). Theorem 3.2 and Corollary 3.3 is a population statement: if the full GP law were known (equivalently, the κ-indexed kernel family), then the unimodular diffusion geometry is uniquely determined. In practice we estimate parameters from finitely many observations at a chosen (possibly learned) κ, so geometry recovery is statistical and regularisation-dependent. The unimodular gauge removes an exact nuisance direction in which an invariant density can improve likelihood without changing the leading decay distance (Remark 3.13); empirically, this confounding is visible in GLSEA, where the density-augmented variant improves predictive metrics but degrades the geometry diagnostic (Table 1).

Remark 3.5 (Constructive inversion (Post–Widder)). Theorem 3.2 is constructive. Writing $s = \kappa^2$ and $F_{x,y}(s) := \frac{\Gamma(\alpha)}{\sigma^2} k(x, y; \sqrt{s}) = \int_0^\infty e^{-st}\, t^{\alpha-1} q_t(x, y)\, \mathrm{d}t$, the Post–Widder inversion formula recovers $q_t(x, y)$ from derivatives of $F_{x,y}(s)$ under mild regularity/growth assumptions; see Appendix D.1.

Geometric content (separate from identifiability). Theorem 3.2 is an inverse statement about the generator. The results below describe the geometry encoded by this generator: a determinant split isolates the unimodular diffusion geometry from an invariant density, and Varadhan asymptotics identify the unimodular geometry as the exponent controlling short-range correlation decay. Throughout, the operator is understood in the same boundary setting as in Section 2.

3.1. Drift–diffusion decomposition and model variants

Proposition 3.6 (Drift–diffusion decomposition)**.** *Let* $\tilde{A} : \Omega \to \mathbb{R}^{d\times d}$ *be an SPD field with sufficient regularity for pointwise determinants and weak derivatives, and assume uniform ellipticity and boundedness: there exist* $0 < a \le b < \infty$ *such that* $aI \preceq \tilde{A}(x) \preceq bI$ *for almost every* $x \in \Omega$*. Define*

$$\begin{aligned} \rho(x) &:= \big(\det \tilde{A}(x)\big)^{\frac{1}{d}}, \\ \bar{A}(x) &:= \rho(x)^{-1}\tilde{A}(x), \qquad \det \bar{A}(x) \equiv 1. \end{aligned} \tag{10}$$

Define the weighted diffusion generator

$$\begin{aligned} \mathcal{L}_{\bar{A},\rho} f &:= \frac{1}{\rho}\, \nabla \cdot (\rho\, \bar{A}\nabla f) \\ &= \nabla \cdot (\bar{A}\nabla f) + \langle \bar{A}\nabla \log \rho, \nabla f\rangle. \end{aligned} \tag{11}$$

Equivalently, $\nabla \cdot (\tilde{A}\nabla f) = \rho\, \mathcal{L}_{\bar{A},\rho} f$ *for smooth* f*. Under reflecting (no-flux) boundary conditions* $\big((\rho\, \bar{A}\nabla f) \cdot n = 0\big)$*,* $\mathcal{L}_{\bar{A},\rho}$ *is symmetric on* $L^2(\Omega, \rho\, \mathrm{d}x)$ *and is the generator of a reversible diffusion with invariant measure* $\rho\, \mathrm{d}x$*. The diffusion term is driven by the unimodular part* $\bar{A}$ *(intrinsic geometry), while* ρ *enters through the weighted divergence and induces the explicit drift* $\bar{A}\nabla \log \rho$ *in* (11).

Definition 3.7 (RDM-Geo (pure geometry))**.** RDM-Geo learns only a unimodular inverse metric field $A(x)$ with $\det A(x) = 1$ almost everywhere and defines $\mathcal{K} = \sigma^2(\kappa^2 - \Delta_g)^{-\alpha}$, where $g = A^{-1}$. Under unimodularity, $\kappa^2 - \Delta_g = \kappa^2 - \nabla \cdot (A\nabla)$.

Definition 3.8 (RDM-Full (reversible diffusion))**.** RDM-Full learns a general SPD diffusion field $\tilde{A}(x)$, writes $\tilde{A} = \rho\bar{A}$ as in Equation (10), and defines the covariance operator by

$$\mathcal{K} = \sigma^2\big(\kappa^2 - \mathcal{L}_{\bar{A},\rho}\big)^{-\alpha},$$

where $\mathcal{L}_{\bar{A},\rho}$ is the reversible-diffusion generator in (11). The operator $\kappa^2 - \mathcal{L}_{\bar{A},\rho}$ is self-adjoint on $L^2(\Omega, \rho\, \mathrm{d}x)$; equivalently, the defining SPDE uses σ times isonormal white noise on this weighted space. When $\alpha > \frac{d}{2}$, $\mathcal{K}$ is trace-class on $L^2(\Omega, \rho\, \mathrm{d}x)$ and admits an integral kernel $k_{\alpha,\rho}$ with respect to $\rho\, \mathrm{d}x$. In this interpretation, $\bar{A}$ specifies the diffusion geometry while ρ is the invariant density (entering through the drift and measure); RDM-Geo is recovered by setting $\rho \equiv 1$.

Remark 3.9 (Weighted variational form). On $L^2(\Omega, \rho\, \mathrm{d}x)$, the operator $\kappa^2 - \mathcal{L}_{\bar{A},\rho}$ is associated with the coercive bilinear form $a_\rho(u, v) = \kappa^2 \int_\Omega uv\, \rho\, \mathrm{d}x + \int_\Omega (\bar{A}\nabla u) \cdot \nabla v\, \rho\, \mathrm{d}x$, so self-adjointness follows exactly as in Section 2.3 but with the weighted inner product $(u, v)_\rho = \int uv\, \rho\, \mathrm{d}x$. Appendix Appendix F discretises a_ρ with FEM and shows how the resulting weighted matrices enter the rSPDE and SLQ inference pipeline.

3.2. Geodesic control of short-range decay

We now connect the learned geometry to the *asymptotic* correlation decay of both RDM-Geo and RDM-Full. The key point is that in the short-time/high-κ regime, the leading exponential decay is governed by the *diffusion tensor* (here the unimodular part $\bar{A}$), while the weight ρ enters only through lower-order effects (drift, measure, and sub-exponential amplitudes).

Definition 3.10 (Correlation geometry)**.** Let $\{k(\cdot, \cdot; \kappa)\}_{\kappa>0}$ be a family of positive-valued kernels. Its *correlation geometry* is the (pseudo-)distance

$$d_k(x, y) := \lim_{\kappa\to\infty} -\frac{1}{\kappa} \log k(x, y; \kappa), \tag{12}$$

whenever the limit exists.

For RDM, $k_{\alpha,\rho}(x, y) > 0$ as a positive mixture of the heat kernel (Equation (18) and Theorem 3.12), so the logarithm in Equation (12) is well-defined; we plot $|\mathrm{corr}|$ only for visual stability.

Proposition 3.11 (Conformal gauge freedom)**.** *Let $\tilde{A}$ be an SPD field and define $(\rho, \bar{A})$ by Equation (10). For any scalar field $c : \Omega \to (0, \infty)$, set $\tilde{A}'(x) = c(x)\tilde{A}(x)$ and define $(\rho', \bar{A}')$ analogously. Then $\bar{A}' \equiv \bar{A}$ and $\rho' = c\rho$; in particular the diffusion geometry $\bar{g} = \bar{A}^{-1}$ is invariant under pointwise conformal rescalings of $\tilde{A}$. Equivalently, $\bar{A}$ is the unique pointwise unimodular representative of the conformal class $\tilde{A} \sim c\tilde{A}$.*

Proof. $\rho' = (\det \tilde{A}')^{1/d} = (c^d \det \tilde{A})^{1/d} = c\rho$, hence $\bar{A}' = (\rho')^{-1}\tilde{A}' = \bar{A}$. □

Local anisotropy (heuristic). If $\bar{A}$ is smooth, freezing coefficients suggests that locally around x_0 the kernel behaves like an anisotropic Matérn with Mahalanobis norm $\|h\|_{\bar{A}(x_0)} = \sqrt{h^\top \bar{A}(x_0)h}$, so the directional correlation length in direction v scales as $\ell_v(x_0) \propto \left(v^\top \bar{A}(x_0)v\right)^{-\frac{1}{2}}$. This is the standard parametrix intuition for variable-coefficient elliptic operators (Gilbarg & Trudinger, 2001).

Theorem 3.12 (Geodesic control of short-range decay (RDM-Geo and RDM-Full))**.** *Assume $\tilde{A} \in C^2(\bar{\Omega}; \mathrm{SPD}(d))$ and define $\rho = (\det \tilde{A})^{\frac{1}{d}}$ and $\bar{A} = \rho^{-1}\tilde{A}$ as in Equation (10), so $\bar{A} \in C^2(\bar{\Omega}; \mathrm{SPD}^1(d))$ and $\rho \in C^2(\bar{\Omega})$ with $\rho > 0$. Let $\mathcal{L}_{\bar{A},\rho}$ be the reversible-diffusion generator (11), and let $q_t^{(\rho)}(x, y)$ denote the heat kernel of $e^{t\mathcal{L}_{\bar{A},\rho}}$ under the boundary convention of Section 2, i.e. the integral kernel with respect to the measure $\rho\ \mathrm{d}x$:*

$$(e^{t\mathcal{L}_{\bar{A},\rho}} f)(x) = \int_\Omega q_t^{(\rho)}(x, y)\, f(y)\, \rho(y)\ \mathrm{d}y.$$

Let $\bar{g} = \bar{A}^{-1}$ and denote by $d_{\bar{g}}$ the induced geodesic distance. Fix $\delta > 0$ and let $x, y \in \Omega$ satisfy $\mathrm{dist}(\{x, y\}, \partial\Omega) \geq \delta$ and $x \neq y$. Then the interior Varadhan asymptotic holds (Varadhan, 1967; Stroock & Varadhan, 1997):

$$-4t\ \log q_t^{(\rho)}(x, y) \ \longrightarrow\ d_{\bar{g}}(x, y)^2, \qquad t \downarrow 0, \tag{13}$$

Moreover, define the Matérnised kernel

$$k_{\alpha,\rho}(x, y) := \frac{\sigma^2}{\Gamma(\alpha)} \int_0^\infty t^{\alpha-1} e^{-\kappa^2 t}\, q_t^{(\rho)}(x, y)\ \mathrm{d}t,$$

which coincides with the RDM-Geo kernel when $\rho \equiv 1$ and $\bar{A} = A = g^{-1}$, and with the RDM-Full kernel (as an integral kernel on $L^2(\Omega, \rho\ \mathrm{d}x)$) for general ρ. Then, at such fixed $x \neq y$,

$$-\frac{1}{\kappa} \log k_{\alpha,\rho}(x, y) \ \longrightarrow\ d_{\bar{g}}(x, y), \qquad \kappa \to \infty, \tag{14}$$

so the leading exponential decay rate in the short-range, high-frequency regime is governed by the unimodular geometry $\bar{g}$. In particular, the weight ρ affects only sub-exponential terms (through the drift and amplitude), while the exponent depends only on $\bar{A}$. The parameter α affects only sub-exponential terms (a polynomial prefactor in κ); see Appendix B for a Laplace-method derivation.

Proof. Varadhan's limit (13) is classical; the derivation of (14) (and a non-asymptotic off-diagonal upper bound) is given in Appendix D.5. □

Remark 3.13 (Gauge fixing for well-posed geometry learning). By Theorem 3.12, the correlation geometry (in the sense of Definition 3.10) depends only on the unimodular diffusion tensor $\bar{A}$ and is invariant to conformal rescalings of $\tilde{A}$ (Proposition 3.11). Thus the determinant factor (equivalently ρ) spans an *exact* nuisance direction for geometry: it changes drift and sub-exponential amplitudes without changing the leading decay distance. Since marginal likelihood is highly sensitive to these sub-exponential terms (through amplitude and the induced drift; Equation (11)), optimisation can improve fit by varying ρ while leaving the target distance unchanged, making geometry learning ill-posed (cf. Figure 1). RDM-Geo fixes this gauge by enforcing $\det A \equiv 1$ pointwise, while RDM-Full keeps $\bar{A}$ unimodular and models the remaining degree of freedom explicitly via ρ.

4. Model and inference

We model nonstationarity through a *learned* unimodular diffusion tensor $A(x) \in \mathrm{SPD}^1(d)$ (equivalently $g = A^{-1}$), which is the intrinsic geometry identifiable from the GP law (equivalently, from the kernel family) under our kernel-family correspondence (Section 3). The prior is defined generator-first by the operator $\mathcal{K} = \sigma^2(\kappa^2 - \Delta_g)^{-\alpha}$, so learning a kernel amounts to learning the diffusion geometry inside Δ_g. Training then reduces to maximum marginal likelihood with respect to the parameters of A_θ, implemented via sparse FEM discretisation and stochastic trace estimation, yielding an end-to-end map from data to the learnable geometry.

4.1. Definition via SPDE and equivalent representations

We define f as the (weak) solution to the geometric SPDE

$$\left(\kappa^2 - \Delta_g\right)^{\frac{\alpha}{2}} f = \sigma\, \mathcal{W}, \qquad \alpha > 0, \tag{15}$$

where Δ_g is the Laplace–Beltrami operator associated with a learned unimodular metric field g (equivalently $A = g^{-1}$ with $\det A \equiv 1$ by Assumption 4.4) and $\mathcal{W}$ denotes isonormal Gaussian white noise on $L^2(\Omega)$.

Definition 4.1 (RDM kernel)**.** The RDM covariance operator is $\mathcal{K} := \sigma^2(\kappa^2 - \Delta_g)^{-\alpha}$. When $\alpha > \frac{d}{2}$, $\mathcal{K}$ is trace-class and admits an integral kernel, which we denote $k_\alpha(x, x') = \sigma^2 G_\alpha(x, x')$. Here G_α is the Green function

of $(\kappa^2 - \Delta_g)^\alpha$ (in the distributional sense):

$$(\kappa^2 - \Delta_g)^\alpha G_\alpha(\cdot, x') = \delta_{x'}, \qquad k_\alpha = \sigma^2 G_\alpha. \tag{16}$$

Under Assumption 4.4, $(\kappa^2 - \Delta_g) = (\kappa^2 - \nabla\cdot(A\nabla)) = L_{A,\kappa}$, so all operator-theoretic statements in Section 2 apply unchanged. By spectral calculus, if $\{(\lambda_j, \varphi_j)\}$ are the eigenpairs of $L_{A,\kappa}$, then

$$k_\alpha(x, x') = \sigma^2 \sum_{j\geq 1} \lambda_j^{-\alpha}\, \varphi_j(x)\varphi_j(x'). \tag{17}$$

Using (6) and denoting by p_t the heat kernel of $L_{A,\kappa}$, we also have the heat-kernel mixture

$$k_\alpha(x, x') = \frac{\sigma^2}{\Gamma(\alpha)} \int_0^\infty t^{\alpha-1}\, p_t(x, x')\, \mathrm{d}t, \tag{18}$$

which connects the construction to diffusion kernels (Kondor & Lafferty, 2002; Lafferty & Lebanon, 2005). When writing $L_{A,\kappa} = \kappa^2 - \mathcal{L}$ with $\mathcal{L} = \Delta_g$ as in Theorem 3.2, the heat kernels satisfy $p_t(x, y) = e^{-\kappa^2 t} q_t(x, y)$; inserting into Equation (18) yields the kernel-family Laplace form Equation (9).

Proposition 4.2 (Process-convolution form)**.** *Assume $\alpha > \frac{d}{2}$ so that $L_{A,\kappa}^{-\frac{\alpha}{2}}$ is Hilbert–Schmidt and admits an $L^2(\Omega\times\Omega)$ kernel $G_{\frac{\alpha}{2}}(\cdot,\cdot)$. Then*

$$k_\alpha(x, x') = \sigma^2 \int_\Omega G_{\frac{\alpha}{2}}(x, s)\, G_{\frac{\alpha}{2}}(x', s)\, \mathrm{d}s. \tag{19}$$

Proof. Deferred to Appendix D.2. □

4.2. Gauge fixing and scale non-identifiability

The pair (κ, A) in $L_{A,\kappa} = \kappa^2 - \nabla\cdot(A\nabla)$ is scale non-identifiable: a scalar rescaling of A can be absorbed by κ and σ without changing the covariance family.

Proposition 4.3 (Scale non-identifiability)**.** *For any $c > 0$, $L_{cA,\sqrt{c}\,\kappa} = c\, L_{A,\kappa}$. Consequently,*

$$\sigma^2 L_{A,\kappa}^{-\alpha} = \left(c^{\frac{\alpha}{2}}\sigma\right)^2 L_{cA,\sqrt{c}\,\kappa}^{-\alpha},$$

so (κ, A, σ) are not jointly identifiable without a gauge constraint.

Proof. Deferred to Appendix D.3. □

Assumption 4.4 (Unimodular gauge (RDM-Geo))**.** RDM-Geo is defined in the pointwise unimodular gauge $\det A(x) = 1$ almost everywhere (equivalently $\det g(x) = 1$ for $g = A^{-1}$), i.e. $\sqrt{|g|}\, \mathrm{d}x = \mathrm{d}x$ in the fixed Euclidean coordinates. This fixes the determinant nuisance direction (see Remark 3.13) and, under sufficient regularity, yields $\Delta_g = \nabla\cdot(A\nabla)$ (cf. Equation (5)).

4.3. Implicit neural parameterisation of the metric field

RDM learns a *field* $A(x) \in \mathrm{SPD}^1(d)$, not a finite parameter table. We therefore use an implicit neural representation (INR) (Tancik et al., 2020; Sitzmann et al., 2020) that maps coordinates x to a log-metric chart and constructs constraints by design. Implementation details of the INR (coordinate normalisation, positional encoding, and the discrete regularisers used in training) are provided in Appendix C. Concretely, we exploit the global diffeomorphism between symmetric trace-free matrices and $\mathrm{SPD}^1(d)$ induced by the matrix exponential and the principal matrix logarithm (see, e.g., Bhatia 2007; Pennec et al. 2006; Arsigny et al. 2006). Let $\tilde{S}_\theta(x) \in \mathbb{R}^{d\times d}$ be the raw network output (after a positional encoding). Define the symmetric trace-free projection

$$\Pi_{\mathrm{sym},0}(M) \;:=\; \tfrac{1}{2}(M + M^\top) - \tfrac{\mathrm{tr}(M)}{d}\, I,$$

and set

$$S_\theta(x) = \Pi_{\mathrm{sym},0}\big(\tilde{S}_\theta(x)\big), \qquad A_\theta(x) = \exp\big(S_\theta(x)\big). \tag{20}$$

Then $A_\theta(x) \succ 0$ and $\det A_\theta(x) = \exp(\mathrm{tr}\, S_\theta(x)) = 1$, so the unimodular gauge holds pointwise.

Geometric context. We summarise background on $\mathrm{SPD}^1(d)$ in Appendix G.

Proposition 4.5 (Global log coordinates and ellipticity control)**.** *Let S be symmetric and trace-free and define $A = \exp(S)$. Then $A \in \mathrm{SPD}^1(d)$ and the map $S \mapsto \exp(S)$ is a diffeomorphism from the space of symmetric trace-free matrices onto $\mathrm{SPD}^1(d)$ with inverse given by the principal matrix logarithm. Moreover, $\lambda_i(A) = \exp(\lambda_i(S))$; hence if $\|S\|_2 \leq s_{\max}$ then*

$$e^{-s_{\max}} I \preceq A \preceq e^{s_{\max}} I, \qquad \mathrm{cond}(A) \leq e^{2s_{\max}}.$$

Proof. Deferred to Appendix D.4. □

Uniform ellipticity as a guardrail. To enforce Assumption 2.1 in training, we cap the log-metric norm (Appendix Appendix C); by Proposition 4.5 this yields the uniform bounds $e^{-s_{\max}} I \preceq A_\theta(x) \preceq e^{s_{\max}} I$ and $\mathrm{cond}(A_\theta(x)) \leq e^{2s_{\max}}$.

Regularisation. Learning a geometric field from finite data is ill-posed, so we regularise the log-field (e.g. L^2 and H^1 penalties; Appendix C). We learn θ by maximising the marginal likelihood (type-II ML) using gradients through the sparse linear-algebra pipeline in Section 4.4.

4.4. Sparse inference and training

After FEM discretisation, learning reduces to sparse assembly and sparse linear algebra: the INR enters only through stiffness assembly by evaluating $A_\theta(x)$ at quadrature points.

We use a cov-rSPDE construction to obtain a sparse GMRF prior for general $\alpha > 0$, compute the marginal likelihood in information form, and estimate log-determinants/traces with Hutchinson–SLQ using only SPD solves (PCG+AMG). Gradients are obtained by differentiating assembly (through A_θ at quadrature points) and using standard trace identities, without backpropagating through unrolled Krylov iterations. Full matrix formulas and derivations are deferred to Appendix Appendix E; the weighted RDM-Full analogue is in Appendix Appendix F.

5. Experiments

We evaluate whether RDM turns nonstationary kernel learning into *geometry learning* in practice via (i) a controlled synthetic problem with known ground-truth geometry, and (ii) a real physical field over a complex, multiply-connected domain. Code and configuration files for all experiments are provided in the supplementary material.

5.1. Synthetic sanity check and ablations

We construct a 2D domain with holes and narrow passages, define a ground-truth unimodular metric $A_\star(x) \in \mathrm{SPD}^1(2)$ with spatially varying anisotropy, sample $f_\star \sim \mathcal{N}(0, (\kappa^2 - \nabla\cdot(A_\star\nabla))^{-\alpha})$, and observe noisy points $y_n = f_\star(x_n) + \varepsilon_n$.

Correlation decay (Figure 1). We plot prior correlation magnitude against the ground-truth geodesic distance $d_{g_\star}$, directly testing Theorem 3.12. RDM-Geo closely tracks the oracle decay; the stationary Matérn baseline ($A \equiv I$) shows systematic mismatch, and dropping the unimodular gauge (det unconstrained) yields a strongly distorted profile due to determinant confounding. Quantitatively, the integrated squared error (ISE) between the oracle and learned binned $\mathbb{E}[|\mathrm{corr}|]$ decay curves is 7.5×10^{-4} (RDM-Geo) vs. 1.4×10^{-2} (Matérn–SPDE) and 2.2×10^{-1} (no-gauge SPD). As shown in the bottom rows of Figure 1, we also observe comparable or distinct profiles for RDM-Full, the nonstationary SPDE of Fuglstad et al. (2015a), and Deep Kernel Learning (Wilson et al., 2016).

5.2. Real data: Great Lakes surface temperature

Dataset and domain. We use the NOAA GLSEA daily lake-surface temperature product (NOAA Great Lakes CoastWatch / GLERL, 2025), focusing on Lake Michigan—a multiply-connected domain with complex shorelines and islands. Preprocessing details (mesh extraction, train/test splits) are in Appendix I.

Baselines. We compare RDM-Geo (our geometry-learning model) and the accuracy-oriented variant RDM-Full against: (i) stationary Matérn–SPDE ($A \equiv I$); (ii) the anisotropic nonstationary SPDE baseline of Fuglstad et al. (2015a); (iii) SVGP-RBF; (iv) DKL (Wilson et al., 2016); (v) spectral mixture (SM); and (vi) a no-gauge SPD ablation (det unconstrained). Operator-based methods respect boundaries by construction; extrinsic baselines can "leak" across land.

Metrics. We report predictive RMSE, negative log predictive density (NLPD), and empirical coverage of nominal 90% predictive intervals (Cov@0.9). For boundary awareness, we match point pairs by Euclidean distance and label them as *unobstructed* (segment in water) or *obstructed* (segment crosses land). A geometrically faithful prior should assign lower correlation to obstructed pairs; we report a correlation separation gap and the AUC of using prior correlation as a score to distinguish obstructed vs. unobstructed pairs.

Results (Table 1). *Geometry:* RDM-Geo yields the strongest boundary-aware geometry (highest Gap/AUC), confirming the unimodular gauge as a key inductive bias; the no-gauge ablation loses most of this gain, and extrinsic baselines are near-chance on the diagnostic. *Prediction:* RDM-Full (reversible diffusion) is best on RMSE/NLPD/coverage, but its geometry diagnostic drops substantially relative to RDM-Geo, consistent with density degrees of freedom absorbing signal that would otherwise be forced into the diffusion geometry. RDM-Geo remains competitive on RMSE/NLPD but under-covers relative to the 0.9 target; SVGP-RBF can achieve low RMSE yet severely under-covers.

Sensitivity. A lightweight sweep of key hyperparameters ($s_{\max}$, regularisation strength, and $(\alpha, \kappa_{\text{mult}})$) is reported in Appendix Appendix H, together with a reproducible tuning recipe.

The implementation code has been uploaded via supplementary material.

6. Discussion

RDM recasts nonstationary kernel learning as geometry learning: a single unimodular metric field $A(x)$ inside a sparse SPDE generator. Validity (PSD) follows from spectral calculus; scalability follows from FEM sparsity. The unimodular gauge resolves a global scale degeneracy and focuses the learned field on anisotropy, while our Varadhan analysis links the induced geodesic distance to correlation decay, making the geometry directly interpretable.

Scope. RDM is designed for low-dimensional spatial domains ($d = 2, 3$) where locality, boundary conditions, and a differential operator prior are natural modelling assumptions. We do not position it as a general-purpose method for high-dimensional manifold learning: generic feature spaces lack a canonical Laplace–Beltrami structure or boundary specification, and comparisons would require additional modelling choices orthogonal to our contributions.

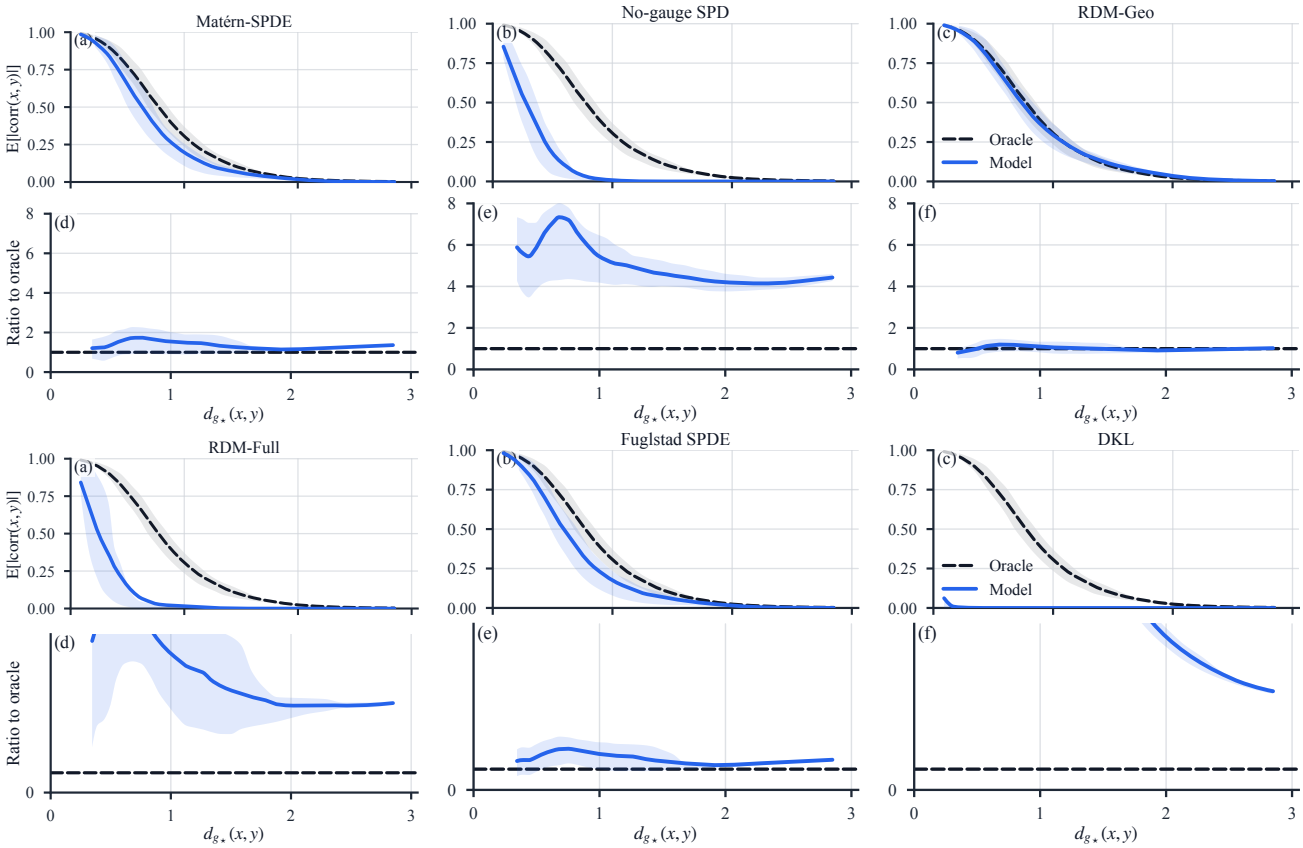

Figure 1. Synthetic: correlation decay as a function of ground-truth geodesic distance $d_{g_\star}(x, y)$. Top two rows: Matérn–SPDE ($A \equiv I$), no-gauge SPD (det unconstrained), and RDM-Geo (unimodular). Bottom two rows: RDM-Full, Fuglstad SPDE, and DKL. For each method, the upper panel shows $\mathbb{E}[|\mathrm{corr}(x, y)|]$ and the lower panel shows the relative log-decay ratio (dashed line at 1). Dashed curves: oracle prior using $A_\star$.

Table 1. GLSEA (Lake Michigan): predictive performance, geometry diagnostic, and compute profile (mean±std over 3 seeds).

Metric	Matérn–SPDE	Fuglstad-SPDE	SVGP-RBF	DKL	SM	**RDM-Geo (ours)**	RDM no-gauge	RDM-Full
RMSE ↓	1.1407±0.0042	1.2358±0.0060	0.8640±0.1883	1.4455±0.2068	1.3032±0.0046	1.2441±0.0083	1.2784±0.0010	**0.8257±0.0544**
NLPD ↓	2.884±0.046	1.931±0.008	1.618±0.507	2.270±0.520	1.752±0.004	1.803±0.012	2.660±0.020	**1.362±0.060**
Cov@0.9	0.727±0.002	0.784±0.006	0.702±0.165	0.851±0.059	0.810±0.008	0.889±0.002	0.751±0.005	**0.950±0.015**
Gap ↑	0.058±0.000	0.095±0.002	0.000±0.002	0.000±0.001	−0.001±0.003	**0.110±0.002**	0.092±0.001	0.048±0.001
AUC ↑	0.549±0.000	0.601±0.003	0.502±0.016	0.514±0.029	0.503±0.017	**0.605±0.002**	0.597±0.000	0.538±0.002
Train(s)	305.4±15.2	335.8±20.9	174.9±3.2	473.2±11.5	216.8±5.6	362.6±21.2	181.8±2.5	1197.5±164.2
Pred(s)	13.9±0.2	14.9±2.4	0.3±0.0	4.7±0.5	0.2±0.0	15.6±1.6	9.0±0.5	64.8±16.2

Limitations and future work. Learning a metric field from finite data is ill-posed and nonconvex, demanding careful regularisation. Our experiments suggest that introducing an invariant density (RDM-Full) can be counterproductive when the goal is to recover boundary-aware geometry, highlighting confounding and the need for better-posed joint priors on $(\bar{A}, \rho)$. The rSPDE and SLQ approximations introduce numerical errors requiring monitored solver tolerances. Future directions include (i) joint priors that allow density variation without degrading diffusion geometry, (ii) geometric analysis of boundary-induced anisotropy and long-range effects, and (iii) GPU-native, end-to-end differentiable pipelines for large-scale geometry learning.

Impact Statement

This work introduces a method for learning intrinsic Riemannian geometry from spatial data through Gaussian processes. RDM's ability to learn anisotropic, boundary-aware correlation structures can improve spatial predictions in environmental monitoring (lake-surface temperature, air quality, ocean currents), climate science, and geophysics—domains where physical boundaries and directional heterogeneity matter. The interpretability of the learned metric

field, where geodesic distance controls correlation decay, provides a transparent link between model structure and physical intuition, potentially informing policy decisions and increasing trust in model-based reasoning.

As with any spatial modelling technique, RDM could be misused for surveillance or predictive policing. The geometry-learning capability could, in principle, infer sensitive spatial patterns from sparse observations. Responsible deployment requires domain-specific ethical review, data governance, and adherence to spatial data ethics guidelines; we encourage practitioners to ensure appropriate consent and anonymisation when working with sensitive location data.

The learned geometry is only as reliable as the data and regularisation allow; in data-sparse regions the prior dominates, and practitioners should validate inferred anisotropy against domain knowledge. The sparse SPDE–FEM formulation reduces memory requirements compared to dense kernel methods, democratising access to nonstationary modelling without high-end GPU clusters. Our experiments run on a single workstation; we advocate for reporting compute budgets in future large-scale extensions.

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A. Related Work

Operator-first versus kernel-first nonstationarity. Traditional nonstationary kernels directly specify a pairwise similarity $k(x, x')$ by modelling endpoint-local lengthscales or covariances (Gibbs, 1997; Paciorek & Schervish, 2003; 2006; Higdon, 1998; Saad et al., 2024). Since the object is a dense covariance rule, sparsity must be recovered through generic approximations. By contrast, RDM learns the generator $\kappa^2 - \Delta_g$ in Equation (15), inheriting both positive-semidefiniteness from spectral calculus and sparsity from the differential operator—the strategy pioneered by Matérn–SPDE methods (Lindgren et al., 2011; 2022). Nonstationary SPDE extensions with spatially varying anisotropy (Fuglstad et al., 2015a) share this viewpoint; RDM adds the unimodular gauge (Assumption 4.4) to decouple diffusion geometry from local scale (determinant) degrees of freedom, addressing a confounding direction closely related to microergodicity phenomena in Matérn estimation (Zhang, 2004; Fuglstad et al., 2015b).

Intrinsic versus latent-space geometry. Warping methods—from spatial deformations (Sampson & Guttorp, 1992) to DKL (Wilson et al., 2016) and deep GPs (de Souza et al., 2023)—model a map ϕ and compose a stationary base kernel $k_0(\phi(x), \phi(x'))$. Geometry then lives in the latent space: geodesics and volume elements in the original domain Ω are not directly interpretable, and without injectivity constraints folding can collapse distant regions (Ober et al., 2021). RDM instead makes nonstationarity intrinsic by parameterising $A(x) \in \mathrm{SPD}^1(d)$ inside the generator, so that $g = A^{-1}$ defines a globally coherent metric field. Manifold Matérn kernels (Borovitskiy et al., 2020; Li et al., 2023) share the Laplace–Beltrami starting point but typically take geometry as given or construct it offline (Fichera et al., 2023); when geometry is learned, it is often inferred in a latent space (Niu et al., 2023; Jørgensen & Hauberg, 2021). RDM learns the metric on the original domain and pairs it with sparse FEM discretisation.

Diffusion kernels and Varadhan asymptotics. The mixture representation Equation (18) links RDM to diffusion kernels (Kondor & Lafferty, 2002; Lafferty & Lebanon, 2005): k_α Matérnises the heat kernel of $L_{A,\kappa}$. Our Varadhan analysis (Theorem 3.12) shows that the unimodular diffusion geometry controls the leading exponential decay while the invariant density enters only through drift and sub-exponential terms (Varadhan, 1967; Stroock & Varadhan, 1997); a non-asymptotic bound follows from classical heat-kernel estimates (Aronson, 1967; Davies, 1989).

B. Technical details for fractional operators

Domains of fractional powers. For $\beta \geq 0$, the domain of $L_{A,\kappa}^{\beta}$ is

$$\mathcal{D}\big(L_{A,\kappa}^{\beta}\big) = \Big\{u \in L^2(\Omega) : \sum_{j\geq 1} \lambda_j^{2\beta} \langle u, \varphi_j\rangle^2 < \infty\Big\}, \tag{21}$$

with $\|L_{A,\kappa}^{\beta} u\|^2 = \sum_j \lambda_j^{2\beta} \langle u, \varphi_j\rangle^2$. For $\alpha > 0$, the operator $L_{A,\kappa}^{-\frac{\alpha}{2}} : L^2(\Omega) \to L^2(\Omega)$ is defined via spectral calculus and $L_{A,\kappa}^{-\alpha} = \big(L_{A,\kappa}^{-\frac{\alpha}{2}}\big)^2$ is trace-class whenever $\alpha > \frac{d}{2}$ by Section 2.4.

Weak solutions to fractional SPDEs. Let $\alpha > 0$. A random distribution f is a weak solution to $L_{A,\kappa}^{\frac{\alpha}{2}} f = \mathcal{W}$ if for every $g \in L^2(\Omega)$,

$$\langle f, g\rangle = \langle \mathcal{W}, L_{A,\kappa}^{-\frac{\alpha}{2}} g\rangle. \tag{22}$$

Equivalently, in the eigenbasis of $L_{A,\kappa}$ we have $\langle f, \varphi_j\rangle = \lambda_j^{-\frac{\alpha}{2}} \xi_j$ with $\xi_j \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$, so $\mathcal{K} = L_{A,\kappa}^{-\alpha}$.

Sub-exponential prefactor in the high-κ regime. In the unimodular/Laplace–Beltrami setting (RDM-Geo, $\rho \equiv 1$ and $\bar{A} = A = g^{-1}$), the interior heat kernel admits the parametrix form

$$q_t(x, y) = (4\pi t)^{-\frac{d}{2}} \exp\Big(-\frac{d_g(x, y)^2}{4t}\Big)\big(a_0(x, y) + \mathcal{O}(t)\big), \qquad t \downarrow 0,$$

with $a_0(x, y) > 0$ smooth for $\mathrm{dist}(\{x, y\}, \partial\Omega) \geq \delta$. Inserting $p_t(x, y) = e^{-\kappa^2 t} q_t(x, y)$ into (18) and applying Laplace's method to the phase $\kappa^2 t + \frac{d_g(x,y)^2}{4t}$ yields the refined asymptotic

$$k_\alpha(x, y) = C_{\alpha,d}(x, y)\, \kappa^{\frac{d}{2}-\alpha} \exp\big(-\kappa\, d_g(x, y)\big)\,(1 + o(1)), \qquad \kappa \to \infty,$$

for a positive amplitude $C_{\alpha,d}(x,y)$ depending on (α, d) and local geometry through $a_0(x,y)$. In particular, the exponential rate is $d_g(x,y)$ while α affects only the polynomial prefactor. For the reversible-diffusion generator in RDM-Full, the corresponding heat kernel admits an analogous small-time expansion with the same exponent $d_{\bar{g}}(x,y)^2/(4t)$ (and ρ entering only through the amplitude), so the same Laplace-method argument yields an asymptotic of the form $k_{\alpha,\rho}(x,y) = C_{\alpha,d}(x,y;\rho,\bar{A})\,\kappa^{\frac{d}{2}-\alpha} e^{-\kappa d_{\bar{g}}(x,y)}(1+o(1))$; see Theorem 3.12 and (Stroock & Varadhan, 1997).

C. INR and regularisation computation details

Coordinate normalisation. We normalise coordinates by the mesh axis-aligned bounding box. Let $b_{\min}, b_{\max} \in \mathbb{R}^d$ be the coordinate-wise minimum/maximum of the mesh vertices. For any physical coordinate $x \in \Omega \subset \mathbb{R}^d$, we define

$$\tilde{x}_i := \frac{x_i - b_{\min,i}}{b_{\max,i} - b_{\min,i}} \in [0,1], \qquad i = 1,\dots,d.$$

In training we evaluate the INR on mesh cell centroids and store $\tilde{x}$ once, optionally caching the positional encoding.

Deterministic Fourier features. Given $\tilde{x} \in \mathbb{R}^d$ and frequencies $\omega_k = 2^k(\pi\,\omega_{\max})$ for $k = 0,\dots,F-1$, we compute a deterministic positional encoding

$$\gamma(\tilde{x}) := \big[\tilde{x},\ \sin(\omega_0\tilde{x}),\ \cos(\omega_0\tilde{x}),\ \dots,\ \sin(\omega_{F-1}\tilde{x}),\ \cos(\omega_{F-1}\tilde{x})\big],$$

where $\sin$ and $\cos$ act elementwise. This produces a $d(1+2F)$-dimensional feature vector when the raw input $\tilde{x}$ is included (and $2Fd$ otherwise).

Unimodular metric INR (RDM-Geo). The network outputs a raw matrix $\tilde{S}_\theta(\tilde{x}) \in \mathbb{R}^{d\times d}$. We form a trace-free symmetric log-metric by

$$S_\theta(\tilde{x}) = \tfrac{1}{2}\big(\tilde{S}_\theta + \tilde{S}_\theta^\top\big) - \tfrac{\operatorname{tr}(\tilde{S}_\theta)}{d} I,$$

apply the smooth Frobenius cap (guardrail for ellipticity)

$$S_\theta(\tilde{x}) \ \leftarrow\ \frac{s_{\max}}{\sqrt{\|S_\theta(\tilde{x})\|_F^2 + s_{\max}^2}}\, S_\theta(\tilde{x}),$$

and set $A_\theta(\tilde{x}) = \exp(S_\theta(\tilde{x}))$. By construction $A_\theta(\tilde{x}) \in \mathrm{SPD}^1(d)$ pointwise. In code, $\exp(\cdot)$ is evaluated in closed form for $d = 2$ and via eigen-decomposition for $d > 2$.

Full diffusion INR (RDM-Full). For the reversible-diffusion model we parameterise $\tilde{A}_\theta = \rho_\theta\,\bar{A}_\theta$ with $\det \bar{A}_\theta \equiv 1$. Starting from a symmetric $S_\theta(\tilde{x})$, we split trace and trace-free parts:

$$s_\theta(\tilde{x}) = \tfrac{1}{d}\operatorname{tr} S_\theta(\tilde{x}), \qquad S_{\theta,0}(\tilde{x}) = S_\theta(\tilde{x}) - s_\theta(\tilde{x}) I.$$

We compute $\bar{A}_\theta(\tilde{x}) = \exp(S_{\theta,0}(\tilde{x}))$ (with the same Frobenius cap as above) and

$$\rho_\theta(\tilde{x}) = \exp(\operatorname{clip}(s_\theta(\tilde{x}), [-\log\rho_{\max}, \log\rho_{\max}])),$$

then set $\tilde{A}_\theta(\tilde{x}) = \rho_\theta(\tilde{x})\bar{A}_\theta(\tilde{x})$. The trace clamp is an implementation guardrail that prevents extreme density variation during training.

Discrete regularisers. We evaluate the INR on mesh cell centroids and treat the resulting log-field as cellwise constant (DG0). Writing $|T_c|$ for the cell area/volume and S_c for the cellwise log-matrix, we approximate the continuous penalties by

$$\mathcal{R}_{\mathrm{L2}} \approx \sum_c |T_c|\,\|S_c\|_F^2, \qquad \mathcal{R}_\nabla \approx \sum_{(c,c')\in\mathcal{E}} w_{cc'}\,\|S_c - S_{c'}\|_F^2, \quad w_{cc'} = \tfrac{1}{2}\big(|T_c| + |T_{c'}|\big),$$

where $\mathcal{E}$ is the set of neighbouring cell pairs that share an edge/facet. Optionally, we normalise both terms by the total measure $\sum_c |T_c|$ (`reg_normalize`) so that regularisation weights are less sensitive to domain scale. In RDM-Geo we regularise the trace-free S_θ; in RDM-Full we regularise the full symmetric log-matrix $S_\theta = S_{\theta,0} + s_\theta I$ (jointly regularising geometry and density).

D. Deferred proofs

D.1. Kernel-family identifiability and inversion

Proof of Theorem 3.2 and Corollary 3.3. We work with RDM-Geo under the unimodular gauge so that $\mathcal{L} = \Delta_g = \nabla\cdot(A\nabla)$ with reflecting boundary conditions. Let $q_t(x,y)$ denote the heat kernel of $e^{t\mathcal{L}}$ with respect to Lebesgue measure, i.e. $(e^{t\mathcal{L}}f)(x) = \int_\Omega q_t(x,y)f(y)\,\mathrm{d}y$. Define the positive operator $L_\kappa := \kappa^2 - \mathcal{L}$; then $e^{-tL_\kappa} = e^{-\kappa^2 t}e^{t\mathcal{L}}$ and the heat kernels satisfy $p_t(x,y) = e^{-\kappa^2 t}q_t(x,y)$.

Kernel family is a Laplace transform. By the Balakrishnan formula (6) applied to L_κ,

$$L_\kappa^{-\alpha} = \frac{1}{\Gamma(\alpha)}\int_0^\infty t^{\alpha-1}e^{-tL_\kappa}\,\mathrm{d}t.$$

Writing the operator as an integral operator with kernel p_t and using $p_t = e^{-\kappa^2 t}q_t$ yields (9).

Injectivity: kernel family ⇒ heat kernel. Fix $x, y \in \Omega$ and write $s = \kappa^2$. Define

$$f_{x,y}(t) := t^{\alpha-1}q_t(x,y), \qquad F_{x,y}(s) := \frac{\Gamma(\alpha)}{\sigma^2}\,k(x,y;\sqrt{s}) = \int_0^\infty e^{-st}\,f_{x,y}(t)\,\mathrm{d}t.$$

Under the assumed smoothness and uniform ellipticity, q_t is continuous for $t > 0$ and satisfies standard Gaussian bounds; in particular, for fixed x, y the function $f_{x,y}$ is locally integrable on $(0,\infty)$ and of exponential order. Since the Laplace transform is injective on this class (e.g. Widder 1941), if two models satisfy $k_1(x,y;\kappa) = k_2(x,y;\kappa)$ for all $\kappa > 0$ then $F^{(1)}_{x,y} = F^{(2)}_{x,y}$ for all $s > 0$, hence $f^{(1)}_{x,y} = f^{(2)}_{x,y}$ for almost every $t > 0$. Continuity of $q_t(x,y)$ for $t > 0$ upgrades this to pointwise equality $q^{(1)}_t(x,y) = q^{(2)}_t(x,y)$ for all $t > 0$.

Heat kernel ⇒ generator. Equality of q_t implies equality of the semigroups: for every $t > 0$ and admissible test function f,

$$(e^{t\mathcal{L}_1}f)(x) = \int_\Omega q_t(x,y)f(y)\,\mathrm{d}y,$$

and the same formula holds for $\mathcal{L}_2$. Since the generator of a strongly continuous semigroup is uniquely determined by $T_t = e^{t\mathcal{L}}$ via $\mathcal{L}f = \lim_{t\downarrow 0}(T_t f - f)/t$ on its domain (e.g. Pazy 1983), we obtain $\mathcal{L}_1 = \mathcal{L}_2$.

Generator ⇒ diffusion tensor (Dirichlet form identifiability). For RDM-Geo, $\mathcal{L} = \nabla\cdot(A\nabla)$ with reflecting boundary conditions, and the associated Dirichlet form on $H^1(\Omega)$ is

$$\mathcal{E}_A(u,v) := -(\mathcal{L}u, v) = \int_\Omega (A\nabla u)\cdot\nabla v\,\mathrm{d}x.$$

If $\mathcal{L}_1 = \mathcal{L}_2$ then $\mathcal{E}_{A_1} = \mathcal{E}_{A_2}$, hence for $B := A_1 - A_2$ we have $\int_\Omega (B\nabla u)\cdot\nabla v\,\mathrm{d}x = 0$ for all $u, v \in H^1(\Omega)$. Taking $v = u$ gives

$$\int_\Omega (\nabla u)^\top B(x)\,\nabla u\,\mathrm{d}x = 0 \qquad \forall u \in H^1(\Omega). \tag{23}$$

Fix $\xi \in \mathbb{R}^d$ and a Lebesgue point x_0 of B. Let $\phi \in C_c^\infty(\mathbb{R}^d)$ satisfy $\phi \equiv 1$ on $B_{1/2}(0)$ and $\mathrm{supp}(\phi) \subset B_1(0)$, and define $u_r(x) = \xi\cdot(x - x_0)\,\phi((x-x_0)/r)$. Then $u_r \in H^1(\Omega)$ for all sufficiently small r and $\nabla u_r = \xi$ on $B_{r/2}(x_0)$ while $\|\nabla u_r\|$ is uniformly bounded on $B_r(x_0)$. Inserting u_r into (23) and letting $r \downarrow 0$ yields

$$\xi^\top B(x_0)\,\xi = 0.$$

Since ξ is arbitrary and this holds for almost every Lebesgue point x_0, we conclude $B = 0$ a.e., i.e. $A_1 = A_2$ a.e. (and thus $g_1 = g_2$ a.e. under unimodularity). □

D.2. Process-convolution form

Proof of Proposition 4.2. Let $T := L_{A,\kappa}^{-\frac{\alpha}{2}}$. Since $L_{A,\kappa}$ is positive and self-adjoint on $L^2(\Omega)$, T is also self-adjoint by functional calculus. By assumption, T is Hilbert–Schmidt, hence there exists a kernel $G_{\frac{\alpha}{2}} \in L^2(\Omega\times\Omega)$ such that for all $f \in L^2(\Omega)$,

$$(Tf)(x) = \int_\Omega G_{\frac{\alpha}{2}}(x,s)\,f(s)\,\mathrm{d}s \qquad \text{for a.e. } x \in \Omega.$$

Moreover, self-adjointness implies $G_{\frac{\alpha}{2}}(x,s) = G_{\frac{\alpha}{2}}(s,x)$ for almost every (x,s) (up to modification on a null set).

The covariance operator is $\mathcal{K} = \sigma^2 L_{A,\kappa}^{-\alpha} = \sigma^2 T^2$. For $f \in L^2(\Omega)$ and a.e. x,

$$\begin{aligned}(\mathcal{K}f)(x) &= \sigma^2\,(T(Tf))(x) = \sigma^2 \int_\Omega G_{\frac{\alpha}{2}}(x,s)\,(Tf)(s)\ \mathrm{d}s \\ &= \sigma^2 \int_\Omega G_{\frac{\alpha}{2}}(x,s) \left(\int_\Omega G_{\frac{\alpha}{2}}(s,x')\,f(x')\ \mathrm{d}x' \right) \mathrm{d}s \\ &= \sigma^2 \int_\Omega \left(\int_\Omega G_{\frac{\alpha}{2}}(x,s)\,G_{\frac{\alpha}{2}}(s,x')\ \mathrm{d}s \right) f(x')\ \mathrm{d}x',\end{aligned}$$

where Fubini is justified by Cauchy–Schwarz since $G_{\frac{\alpha}{2}}(x,\cdot), G_{\frac{\alpha}{2}}(x',\cdot) \in L^2(\Omega)$ for a.e. x, x'. Using symmetry of $G_{\frac{\alpha}{2}}$ gives

$$\int_\Omega G_{\frac{\alpha}{2}}(x,s)\,G_{\frac{\alpha}{2}}(s,x')\ \mathrm{d}s = \int_\Omega G_{\frac{\alpha}{2}}(x,s)\,G_{\frac{\alpha}{2}}(x',s)\ \mathrm{d}s,$$

which is exactly (19). □

D.3. Scale non-identifiability

Proof of Proposition 4.3. Recall the bilinear form a from (2). For any $c > 0$ and $u, v \in H^1(\Omega)$,

$$a_{cA,\sqrt{c}\,\kappa}(u,v) = (\sqrt{c}\,\kappa)^2(u,v) + \int_\Omega (cA\nabla u)\cdot\nabla v\ \mathrm{d}x = c\,a_{A,\kappa}(u,v).$$

Let $L_{A,\kappa}$ and $L_{cA,\sqrt{c}\,\kappa}$ be the self-adjoint operators associated with $a_{A,\kappa}$ and $a_{cA,\sqrt{c}\,\kappa}$ via the Friedrichs extension. The identity above implies $L_{cA,\sqrt{c}\,\kappa} = c\,L_{A,\kappa}$ as operators on $L^2(\Omega)$. By functional calculus for positive self-adjoint operators,

$$L_{cA,\sqrt{c}\,\kappa}^{-\alpha} = (cL_{A,\kappa})^{-\alpha} = c^{-\alpha} L_{A,\kappa}^{-\alpha}.$$

Multiplying by σ^2 gives

$$\sigma^2 L_{A,\kappa}^{-\alpha} = (c^{\frac{\alpha}{2}}\sigma)^2\, L_{cA,\sqrt{c}\,\kappa}^{-\alpha},$$

which shows the stated scale non-identifiability. □

D.4. Global log coordinates and ellipticity control

Proof of Proposition 4.5. Let $\mathrm{Sym}_0(d) = \{S \in \mathbb{R}^{d\times d} : S = S^\top,\ \mathrm{tr}\,S = 0\}$. If $S \in \mathrm{Sym}_0(d)$ then $\exp(S) \in \mathrm{SPD}(d)$ and $\det(\exp(S)) = \exp(\mathrm{tr}\,S) = 1$, hence $A = \exp(S) \in \mathrm{SPD}^1(d)$. Conversely, if $A \in \mathrm{SPD}^1(d)$ then the principal logarithm $\log(A)$ is symmetric and satisfies $\mathrm{tr}(\log A) = \log\det(A) = 0$, hence $\log(A) \in \mathrm{Sym}_0(d)$. Since $\exp : \mathrm{Sym}(d) \to \mathrm{SPD}(d)$ is a smooth bijection with smooth inverse $\log$ (principal logarithm), the restriction $\exp : \mathrm{Sym}_0(d) \to \mathrm{SPD}^1(d)$ is a diffeomorphism with inverse $\log$.

For the eigenvalue identities, diagonalise $S = U\mathrm{diag}(\lambda_1,\dots,\lambda_d)U^\top$ with U orthogonal. Then

$$A = \exp(S) = U\,\mathrm{diag}(e^{\lambda_1},\dots,e^{\lambda_d})\,U^\top,$$

so $\lambda_i(A) = e^{\lambda_i(S)}$. If $\|S\|_2 \le s_{\max}$ then every eigenvalue satisfies $|\lambda_i(S)| \le s_{\max}$, hence $e^{-s_{\max}} \le \lambda_i(A) \le e^{s_{\max}}$ and therefore $e^{-s_{\max}} I \preceq A \preceq e^{s_{\max}} I$. The condition-number bound follows from $\mathrm{cond}(A) = \lambda_{\max}(A)/\lambda_{\min}(A) \le e^{2s_{\max}}$. □

D.5. Geodesic control of short-range decay

Decay limit and off-diagonal bound in Theorem 3.12. Fix $x \neq y$ with $\mathrm{dist}(\{x,y\},\partial\Omega) \ge \delta$ and write $d := d_{\bar{g}}(x,y) > 0$. Let $q_t^{(\rho)}(x,y)$ denote the heat kernel of $e^{t\mathcal{L}_{\bar{A},\rho}}$ with respect to the measure $\rho\ \mathrm{d}x$, as in Theorem 3.12. By definition,

$$k_{\alpha,\rho}(x,y) = \frac{\sigma^2}{\Gamma(\alpha)} \int_0^\infty t^{\alpha-1} e^{-\kappa^2 t}\, q_t^{(\rho)}(x,y)\ \mathrm{d}t.$$

Limit (14)**.** By Varadhan's asymptotic (13), for any $\varepsilon \in (0, d^2)$ there exists $t_\varepsilon > 0$ such that for all $t \in (0, t_\varepsilon)$,

$$\exp\Big(-\frac{d^2+\varepsilon}{4t}\Big) \;\le\; q_t^{(\rho)}(x,y) \;\le\; \exp\Big(-\frac{d^2-\varepsilon}{4t}\Big). \tag{24}$$

Split the defining integral at t_ε. Since $q_t^{(\rho)}(x,y)$ is continuous for $t > 0$ under the stated regularity, it is bounded above on $[t_\varepsilon, \infty)$ by some $C_\varepsilon < \infty$, and therefore

$$\int_{t_\varepsilon}^{\infty} t^{\alpha-1} e^{-\kappa^2 t}\, q_t^{(\rho)}(x,y)\, \mathrm{d}t \le C_\varepsilon \int_{t_\varepsilon}^{\infty} t^{\alpha-1} e^{-\kappa^2 t}\, \mathrm{d}t \le C'_\varepsilon\, e^{-\kappa^2 t_\varepsilon},$$

which is super-exponentially small compared to $e^{-\kappa}$ and hence does not affect the limit of $-\kappa^{-1}\log k_{\alpha,\rho}(x,y)$.

For the small-time contribution, use (24) and define $\phi_c(t) := \kappa^2 t + \frac{c}{4t}$ for $c > 0$. The minimiser of ϕ_c over $t > 0$ is $t_c^\star = \frac{\sqrt{c}}{2\kappa}$, and $\min_{t>0}\phi_c(t) = \kappa\sqrt{c}$. With the change of variables $t = t_c^\star s$,

$$\int_0^{t_\varepsilon} t^{\alpha-1} e^{-\phi_c(t)}\, \mathrm{d}t = (t_c^\star)^\alpha \int_0^{t_\varepsilon/t_c^\star} s^{\alpha-1} \exp\Big(-\frac{\kappa\sqrt{c}}{2}\Big(s+\frac{1}{s}\Big)\Big)\, \mathrm{d}s.$$

Since $\frac{1}{2}(s+s^{-1}) \ge 1$ with equality only at $s = 1$, this integral is bounded above by $C\,\kappa^{-\alpha}e^{-\kappa\sqrt{c}}$ and bounded below by $c'\,\kappa^{-\alpha}e^{-\kappa\sqrt{c}}$ up to factors of the form $e^{\pm\eta\kappa}$ that can be made arbitrarily small by restricting to a sufficiently small neighbourhood of $s = 1$. Applying these bounds with $c = d^2 - \varepsilon$ (upper) and $c = d^2 + \varepsilon$ (lower), and absorbing constants and polynomial factors into $\mathcal{O}(\log\kappa)$ terms, yields

$$\sqrt{d^2-\varepsilon} \;\le\; \liminf_{\kappa\to\infty}\Big(-\frac{1}{\kappa}\log k_{\alpha,\rho}(x,y)\Big) \le \limsup_{\kappa\to\infty}\Big(-\frac{1}{\kappa}\log k_{\alpha,\rho}(x,y)\Big) \le \sqrt{d^2+\varepsilon}.$$

Since $\varepsilon > 0$ is arbitrary, sending $\varepsilon \downarrow 0$ gives (14). A refined Laplace-method evaluation of the polynomial prefactor is given in Appendix B.

Non-asymptotic bound (Remark after Theorem 3.12). Assume additionally $\alpha > \frac{d}{2}$ and uniform ellipticity. Gaussian upper bounds for $q_t^{(\rho)}$ imply the existence of constants $C_0, C_1 > 0$ such that for all sufficiently small t,

$$q_t^{(\rho)}(x,y) \;\le\; C_0\, t^{-\frac{d}{2}} \exp\Big(-\frac{d^2}{C_1 t}\Big).$$

Splitting the t-integral at a fixed small time and bounding the remainder by $e^{-\kappa^2(\cdot)}$ yields

$$k_{\alpha,\rho}(x,y) \;\le\; C\int_0^\infty t^{\alpha-1-\frac{d}{2}} \exp\Big(-\kappa^2 t - \frac{d^2}{C_1 t}\Big)\, \mathrm{d}t.$$

With the change of variables $t = (d/\kappa)s$, the right-hand side becomes

$$C\, d^{\alpha-\frac{d}{2}}\, \kappa^{\frac{d}{2}-\alpha} \int_0^\infty s^{\alpha-1-\frac{d}{2}} \exp\Big(-\kappa d\Big(s+\frac{1}{C_1 s}\Big)\Big)\, \mathrm{d}s.$$

For $s \ge 1$, one has $s + \frac{1}{C_1 s} \ge s$, so $\int_1^\infty s^{\alpha-1-\frac{d}{2}} e^{-\kappa d s}\, \mathrm{d}s \le C'(\kappa d)^{-1} e^{-\kappa d}$. For $s \in (0,1]$, one has $s + \frac{1}{C_1 s} \ge \frac{1}{C_1 s}$, so with $r = 1/s$ we obtain $\int_0^1 s^{\alpha-1-\frac{d}{2}} e^{-\kappa d/(C_1 s)}\, \mathrm{d}s \le C''(\kappa d)^{-1} e^{-\kappa d/C_1}$. Combining the two bounds and absorbing $d^{\alpha-\frac{d}{2}}$ into a domain-dependent constant yields

$$k_{\alpha,\rho}(x,y) \;\le\; C\,\kappa^{\frac{d}{2}-\alpha} \exp\big(-c\,\kappa\, d\big),$$

for some $c > 0$, which is the claimed off-diagonal exponential upper bound. □

E. RDM-Geo discretisation and inference

This appendix expands the summary in Section 4.4 and records the FEM/rSPDE matrices, marginal likelihood identities, and gradient estimators used in our implementation for RDM-Geo ($\rho \equiv 1$). The weighted RDM-Full analogue is given in Appendix F.

E.1. Discretisation and rSPDE prior

Let $\{\psi_i\}_{i=1}^N \subset H^1(\Omega)$ be first-order FEM basis functions. Define

$$\begin{aligned}(\mathbf{M})_{ij} &= \int_\Omega \psi_i \psi_j \, \mathrm{d}x, \\ (\mathbf{K}_A)_{ij} &= \int_\Omega (\nabla \psi_i)^\top A_\theta(x) \nabla \psi_j \, \mathrm{d}x,\end{aligned} \tag{25}$$

and the discrete operator

$$\mathbf{L}_\theta = \kappa^2 \mathbf{M} + \mathbf{K}_A. \tag{26}$$

To support general $\alpha > 0$, we use a covariance-based rSPDE approximation (Bolin & Kirchner, 2020; Bolin et al., 2024):

$$\begin{gathered}\alpha = m + \eta, \quad m = \lfloor \alpha \rfloor, \quad \eta \in [0, 1), \\ L_{A,\kappa}^{-\alpha} \approx L_{A,\kappa}^{-m} \left(kI + \sum_{j=1}^{J} r_j (L_{A,\kappa} - p_j I)^{-1} \right), \\ r_j, k > 0, \quad p_j < 0,\end{gathered} \tag{27}$$

For computation we commit to the cov-rSPDE GMRF construction of Bolin et al. (2024). Let the *lumped* mass matrix be

$$\tilde{\mathbf{M}} \coloneqq \mathrm{diag}(\mathbf{M}\mathbf{1}),$$

so $\tilde{\mathbf{M}}$ is diagonal SPD. Define the κ-scaled operator

$$\bar{\mathbf{L}}_\theta \coloneqq \mathbf{K}_A/\kappa^2 + \tilde{\mathbf{M}}, \qquad \mathbf{P}_\theta \coloneqq \tilde{\mathbf{M}}^{-1} \bar{\mathbf{L}}_\theta,$$

and write $\mathbf{P}_\theta^m$ for the m-fold product. The FEM weights are represented as a sum of $(J+1)$ independent components $\mathbf{w} = \sum_{j=0}^{J} \mathbf{w}^{(j)}$ with block-diagonal precision

$$\mathbf{Q}_\theta = \mathrm{diag}\big(\mathbf{Q}_\theta^{(0)}, \mathbf{Q}_\theta^{(1)}, \ldots, \mathbf{Q}_\theta^{(J)}\big), \tag{28}$$

where, for $j = 1, \ldots, J$,

$$\mathbf{Q}_\theta^{(j)} = \kappa^{2\alpha} \, r_j^{-1} \, (\bar{\mathbf{L}}_\theta - p_j \tilde{\mathbf{M}}) \, \mathbf{P}_\theta^m, \qquad p_j < 0,$$

and the residual block is

$$\mathbf{Q}_\theta^{(0)} = \kappa^{2\alpha} \, k^{-1} \times \begin{cases} \tilde{\mathbf{M}}^{-1}, & m = 0, \\ \tilde{\mathbf{M}} \, \mathbf{P}_\theta^m, & m \geq 1. \end{cases}$$

All blocks are SPD, and $(\bar{\mathbf{L}}_\theta - p_j \tilde{\mathbf{M}})$ is a stable shifted solve since $p_j < 0$.

E.2. Marginal likelihood in information form

Let $\mathbf{y} \in \mathbb{R}^n$ be noisy observations with $\mathbf{R} = \sigma_\varepsilon^2 \mathbf{I}$ and interpolation matrix $(\mathbf{H}_0)_{ni} = \psi_i(x_n)$. With the stacked state $\mathbf{w}$ and global scale σ, use

$$\mathbf{H} = \big[\sigma \mathbf{H}_0 \;\; \sigma \mathbf{H}_0 \;\; \cdots \;\; \sigma \mathbf{H}_0\big],$$

so $\mathbf{y} = \mathbf{H}\mathbf{w} + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \mathbf{R})$, $\mathbf{w} \sim \mathcal{N}(0, \mathbf{Q}_\theta^{-1})$. For computation define

$$\mathbf{\Lambda}_\theta \coloneqq \mathbf{Q}_\theta + \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H}, \qquad \mathbf{b} \coloneqq \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{y}.$$

Then, by the Woodbury identity and the matrix determinant lemma, the marginal log-likelihood $\ell(\theta) = \log p(\mathbf{y}|\theta)$ can be written as

$$\ell(\theta) = \tfrac{1}{2} \, \mathbf{b}^\top \mathbf{\Lambda}_\theta^{-1} \mathbf{b} + \tfrac{1}{2} \Big(\log |\mathbf{Q}_\theta| - \log |\mathbf{\Lambda}_\theta| \Big) + \mathrm{const}. \tag{29}$$

The quadratic term uses one solve $\mathbf{\Lambda}_\theta \mathbf{v} = \mathbf{b}$.

E.3. SLQ log-determinants, gradients, and optimisation

We estimate $\log|\mathbf{Q}_\theta|$, $\log|\mathbf{\Lambda}_\theta|$, and trace terms by stochastic Lanczos quadrature with Hutchinson probes (Hutchinson, 1990; Ubaru et al., 2017). All systems are SPD; we use PCG with AMG preconditioning (including the shifted solves implied by Equation (28)). For any parameter θ_i affecting only $\mathbf{Q}_\theta$ (fixed $\mathbf{H}$ and $\mathbf{R}$),

$$\begin{aligned} \frac{\partial \ell}{\partial \theta_i} &= \tfrac{1}{2}\,\mathrm{tr}\Big(\big(\mathbf{Q}_\theta^{-1} - \mathbf{\Lambda}_\theta^{-1}\big)\,\partial_{\theta_i}\mathbf{Q}_\theta\Big) \\ &\quad - \tfrac{1}{2}\,\mathbf{v}^\top(\partial_{\theta_i}\mathbf{Q}_\theta)\,\mathbf{v}, \\ \mathbf{\Lambda}_\theta \mathbf{v} &= \mathbf{b}. \end{aligned} \tag{30}$$

We obtain $\partial_{\theta_i}\mathbf{Q}_\theta$ by differentiating stiffness assembly through $\partial_\theta A_\theta(x_q)$. Training maximises $\ell(\theta)$ with first-order optimisers (e.g. Adam) using the stochastic gradient in (30), without backpropagating through unrolled Krylov iterations. For prediction we use the standard sparse-Gaussian conditioning formulas, requiring additional solves with $\mathbf{\Lambda}_\theta$.

F. RDM-Full discretisation and inference

This appendix details the finite element (FEM) discretisation for RDM-Full (Definition 3.8) and shows how it fits into the same sparse rSPDE and SLQ pipeline as RDM-Geo (Appendix E). The only conceptual change relative to RDM-Geo is that the generator is self-adjoint on the *weighted* space $L^2(\Omega, \rho\,\mathrm{d}x)$ (a reversible diffusion with invariant measure $\rho\,\mathrm{d}x$), which induces weighted mass and stiffness matrices. Discretising the symmetric weighted variational form keeps all linear systems SPD and avoids separately handling the drift term in the strong form.

F.1. Weighted operator and variational form

Recall $\tilde{A} = \rho\,\bar{A}$ with $\rho = (\det\tilde{A})^{\frac{1}{d}}$ and $\det\bar{A} \equiv 1$ (Equation (10)). The weighted diffusion generator is

$$\mathcal{L}_{\bar{A},\rho} f = \frac{1}{\rho}\,\nabla\cdot(\rho\,\bar{A}\nabla f),$$

and RDM-Full uses the positive self-adjoint operator

$$L_{\bar{A},\rho,\kappa} \;:=\; \kappa^2 - \mathcal{L}_{\bar{A},\rho}$$

on $L^2(\Omega, \rho\,\mathrm{d}x)$. Introduce the weighted inner product $(u,v)_\rho = \int_\Omega uv\,\rho\,\mathrm{d}x$. Under the boundary convention of Section 2, integration by parts yields the symmetric bilinear form

$$a_\rho(u,v) := \kappa^2\int_\Omega uv\,\rho\,\mathrm{d}x + \int_\Omega (\bar{A}\nabla u)\cdot\nabla v\,\rho\,\mathrm{d}x \qquad \text{so that} \qquad a_\rho(u,v) = (L_{\bar{A},\rho,\kappa}u, v)_\rho. \tag{31}$$

This weighted variational form is the correct object to discretise: it makes self-adjointness and positive definiteness explicit (cf. Proposition 3.6) and it implicitly accounts for the drift $\bar{A}\nabla\log\rho$ present in the strong form (11).

F.2. A practical parameterisation of $(\rho, \bar{A})$

To learn $\tilde{A}(x) \in \mathrm{SPD}(d)$ while retaining the determinant split, one convenient choice is to let the INR output an unconstrained matrix field $\tilde{S}_\theta(x) \in \mathbb{R}^{d\times d}$ and symmetrise it:

$$S_\theta(x) = \tfrac{1}{2}\big(\tilde{S}_\theta(x) + \tilde{S}_\theta(x)^\top\big), \qquad s_\theta(x) = \tfrac{1}{d}\mathrm{tr}\,S_\theta(x), \qquad S_{\theta,0}(x) = S_\theta(x) - s_\theta(x)I.$$

Then define

$$\rho_\theta(x) = \exp(s_\theta(x)), \qquad \bar{A}_\theta(x) = \exp\big(S_{\theta,0}(x)\big), \qquad \tilde{A}_\theta(x) = \rho_\theta(x)\,\bar{A}_\theta(x). \tag{32}$$

By construction $\rho_\theta > 0$ and $\det\bar{A}_\theta \equiv 1$, and $\tilde{A}_\theta$ is SPD. This parameterisation is a strict extension of RDM-Geo: setting $s_\theta \equiv 0$ recovers $\rho \equiv 1$ and $\tilde{A} = \bar{A}$.

F.3. Weighted FEM matrices

Let $\{\psi_i\}_{i=1}^N \subset H^1(\Omega)$ be the same FEM basis as in Appendix E. Discretising (31) gives weighted mass and stiffness matrices

$$\begin{aligned}(\mathbf{M}_\rho)_{ij} &= \int_\Omega \psi_i(x)\psi_j(x)\,\rho_\theta(x)\;\mathrm{d}x,\\ (\mathbf{K}_{\bar{A},\rho})_{ij} &= \int_\Omega (\nabla\psi_i(x))^\top \bar{A}_\theta(x)\nabla\psi_j(x)\,\rho_\theta(x)\;\mathrm{d}x,\end{aligned} \tag{33}$$

and the discrete operator

$$\mathbf{L}_{\theta,\rho} = \kappa^2\mathbf{M}_\rho + \mathbf{K}_{\bar{A},\rho}. \tag{34}$$

The matrices in (33) are assembled exactly like Appendix E, except that every quadrature contribution is additionally weighted by $\rho_\theta(x_q)$ and uses $\bar{A}_\theta(x_q)$ rather than $A_\theta(x_q)$. Importantly, assembling via (31) keeps $\mathbf{L}_{\theta,\rho}$ symmetric positive definite; the drift term never appears explicitly.

F.4. rSPDE approximation and block precision

For general $\alpha > 0$ we use the same covariance-based rational approximation as in Equation (27), now applied to $L_{\bar{A},\rho,\kappa}$:

$$L_{\bar{A},\rho,\kappa}^{-\alpha} \approx L_{\bar{A},\rho,\kappa}^{-m}\left(k\,I + \sum_{j=1}^{J} r_j\,(L_{\bar{A},\rho,\kappa} - p_j I)^{-1}\right), \qquad m = \lfloor\alpha\rfloor,\; r_j, k > 0,\; p_j < 0.$$

Under FEM discretisation we use the same cov-rSPDE GMRF construction as in Appendix E. Let $\tilde{\mathbf{M}}_\rho = \mathrm{diag}(\mathbf{M}_\rho\mathbf{1})$ be the lumped weighted mass and define $\bar{\mathbf{L}}_{\theta,\rho} = \mathbf{K}_{\bar{A},\rho}/\kappa^2 + \tilde{\mathbf{M}}_\rho$ and $\mathbf{P}_{\theta,\rho} = \tilde{\mathbf{M}}_\rho^{-1}\bar{\mathbf{L}}_{\theta,\rho}$. Stacking the $(J+1)$ latent components yields a block-diagonal precision

$$\mathbf{Q}_{\theta,\rho} = \mathrm{diag}\big(\mathbf{Q}_{\theta,\rho}^{(0)}, \mathbf{Q}_{\theta,\rho}^{(1)}, \ldots, \mathbf{Q}_{\theta,\rho}^{(J)}\big). \tag{35}$$

The blocks have the same closed form as Equation (28) with $(\tilde{\mathbf{M}}, \bar{\mathbf{L}}_\theta, \mathbf{P}_\theta)$ replaced by $(\tilde{\mathbf{M}}_\rho, \bar{\mathbf{L}}_{\theta,\rho}, \mathbf{P}_{\theta,\rho})$: for $j = 1, \ldots, J$,

$$\mathbf{Q}_{\theta,\rho}^{(j)} = \kappa^{2\alpha}\, r_j^{-1}\,(\bar{\mathbf{L}}_{\theta,\rho} - p_j\tilde{\mathbf{M}}_\rho)\,\mathbf{P}_{\theta,\rho}^m, \qquad p_j < 0,$$

and

$$\mathbf{Q}_{\theta,\rho}^{(0)} = \kappa^{2\alpha}\,k^{-1} \times \begin{cases}\tilde{\mathbf{M}}_\rho^{-1}, & m = 0,\\ \tilde{\mathbf{M}}_\rho\,\mathbf{P}_{\theta,\rho}^m, & m \geq 1.\end{cases}$$

F.5. Observation model and likelihood

Point observations are unchanged: with interpolation matrix $(\mathbf{H}_0)_{ni} = \psi_i(x_n)$ and global scale σ, define $\mathbf{H}$ exactly as in Appendix E and use $\mathbf{y} = \mathbf{H}\mathbf{w} + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \mathbf{R})$. All likelihood identities in Appendix E remain valid with $\mathbf{Q}_\theta$ replaced by $\mathbf{Q}_{\theta,\rho}$ (and hence $\mathbf{\Lambda}_\theta$ replaced by $\mathbf{\Lambda}_{\theta,\rho} = \mathbf{Q}_{\theta,\rho} + \mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}$).

F.6. Gradients through weighted assembly

The gradient identity (30) is unchanged. What changes is $\partial_{\theta_i}\mathbf{Q}_{\theta,\rho}$, which depends on $\partial_{\theta_i}\mathbf{L}_{\theta,\rho}$ and $\partial_{\theta_i}\mathbf{M}_\rho$. From (33) one has, entrywise,

$$\partial_{\theta_i}(\mathbf{M}_\rho)_{ij} = \int_\Omega \psi_i\psi_j\,\partial_{\theta_i}\rho_\theta\;\mathrm{d}x, \qquad \partial_{\theta_i}(\mathbf{K}_{\bar{A},\rho})_{ij} = \int_\Omega (\nabla\psi_i)^\top\Big(\rho_\theta\,\partial_{\theta_i}\bar{A}_\theta + (\partial_{\theta_i}\rho_\theta)\,\bar{A}_\theta\Big)\nabla\psi_j\;\mathrm{d}x,$$

and therefore

$$\partial_{\theta_i}\mathbf{L}_{\theta,\rho} = \kappa^2\,\partial_{\theta_i}\mathbf{M}_\rho + \partial_{\theta_i}\mathbf{K}_{\bar{A},\rho}.$$

Under the parameterisation (32), $\partial_{\theta_i}\rho_\theta = \rho_\theta\,\partial_{\theta_i}s_\theta$ and $\partial_{\theta_i}\bar{A}_\theta$ is the Fréchet derivative of the matrix exponential at $S_{\theta,0}$ applied to $\partial_{\theta_i}S_{\theta,0}$. In practice these derivatives are obtained by automatic differentiation through the quadrature assembly and the matrix exponential.

G. Geometry of $\mathrm{SPD}^1(d)$

Symmetric-space viewpoint. Equipped with the canonical affine-invariant metric, $\mathrm{SPD}^1(d)$ is isomorphic to the homogeneous space $SL(d)$ modulo $SO(d)$ and is a Riemannian symmetric space (Bhatia, 2007; Helgason, 2001). The Cartan decomposition $\mathfrak{sl}(d) = \mathfrak{so}(d) \oplus \mathfrak{p}$ identifies the tangent space at the identity coset with $\mathfrak{p} = \{S = S^\top,\ \mathrm{tr}\, S = 0\}$ (Helgason, 2001). The log-chart $A = \exp(S)$ with $S \in \mathfrak{p}$ coincides with the Riemannian exponential at the identity under the affine-invariant metric (Pennec et al., 2006; Bhatia, 2007).

Proposition G.1 (Hadamard geometry of $\mathrm{SPD}^1(d)$)**.** *Equip $\mathrm{SPD}^1(d)$ with the canonical affine-invariant metric*

$$\langle U, V\rangle_A := \mathrm{tr}\big(A^{-1}UA^{-1}V\big),$$
$$A \in \mathrm{SPD}^1(d),\ \ U, V \in T_A\mathrm{SPD}^1(d).$$

Then $\mathrm{SPD}^1(d)$, viewed as the homogeneous space $SL(d)$ modulo $SO(d)$, is complete, simply connected, and has non-positive sectional curvature. In particular:

(i) for any $A, B \in \mathrm{SPD}^1(d)$ there is a unique minimizing geodesic joining them;

(ii) the Riemannian exponential and logarithm maps are globally well-defined (no cut locus);

(iii) for fixed B, the function $A \mapsto d(A, B)^2$ is geodesically convex.

Proof. We verify the three Hadamard properties (complete, simply connected, non-positive sectional curvature). The three listed consequences then follow from standard theorems for Hadamard manifolds (Cartan–Hadamard and geodesic convexity of squared distance) (Helgason, 2001).

(1) Simply connected. By Proposition 4.5, the principal logarithm gives a global diffeomorphism between $\mathrm{SPD}^1(d)$ and the vector space of symmetric trace-free matrices. Hence $\mathrm{SPD}^1(d)$ is diffeomorphic to $\mathbb{R}^m$ ($m = \frac{d(d+1)}{2} - 1$) and therefore simply connected.

(2) Completeness. First note that $T_A\mathrm{SPD}^1(d) = \{U = U^\top : \mathrm{tr}(A^{-1}U) = 0\}$, since $\frac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0} \det(A + tU) = \det(A)\,\mathrm{tr}(A^{-1}U)$. The affine-invariant metric is invariant under congruences: for any $G \in SL(d)$, the map $\Phi_G(A) = GAG^\top$ satisfies

$$\langle (\Phi_G)_* U, (\Phi_G)_* V\rangle_{\Phi_G(A)} = \langle U, V\rangle_A, \qquad (\Phi_G)_* U = GUG^\top,$$

by cyclicity of the trace. Geodesics of the affine-invariant metric on $\mathrm{SPD}(d)$ are therefore obtained by reducing to the identity; explicitly, for $A \in \mathrm{SPD}^1(d)$ and $U \in T_A\mathrm{SPD}^1(d)$, set $X = A^{-\frac{1}{2}}UA^{-\frac{1}{2}}$ and define

$$\gamma(t) = A^{\frac{1}{2}} \exp(tX) A^{\frac{1}{2}}.$$

This curve is the affine-invariant geodesic with $\gamma(0) = A$ and $\dot\gamma(0) = U$ (Bhatia, 2007; Pennec et al., 2006). Moreover,

$$\det \gamma(t) = \det(A)\det(\exp(tX)) = \exp\big(t\,\mathrm{tr}(X)\big) = \exp\big(t\,\mathrm{tr}(A^{-1}U)\big) = 1,$$

so $\gamma(t) \in \mathrm{SPD}^1(d)$ for all $t \in \mathbb{R}$. Thus $\mathrm{SPD}^1(d)$ is geodesically complete, and hence complete as a metric space.

(3) Non-positive sectional curvature. Identifying $\mathrm{SPD}^1(d)$ with the homogeneous space $SL(d)$ modulo $SO(d)$, the Lie algebra splits as $\mathfrak{sl}(d) = \mathfrak{so}(d) \oplus \mathfrak{p}$ with $\mathfrak{p} = \{S = S^\top, \mathrm{tr}\, S = 0\}$ and $[\,\mathfrak{p}, \mathfrak{p}\,] \subset \mathfrak{so}(d)$. Let B be the Killing form on $\mathfrak{sl}(d)$; for $\mathfrak{sl}(d)$ one has $B(X, Y) = 2d\,\mathrm{tr}(XY)$, so B is positive definite on $\mathfrak{p}$ and negative definite on $\mathfrak{so}(d)$. Equipping this homogeneous space with the $SL(d)$-invariant metric induced by $B|_{\mathfrak{p}}$ (which matches the affine-invariant metric above up to a constant scaling), the curvature tensor at the identity coset satisfies (Helgason, 2001)

$$R(X, Y)Z = -[[X, Y], Z], \qquad X, Y, Z \in \mathfrak{p}.$$

Therefore, for $X, Y \in \mathfrak{p}$,

$$\langle R(X, Y)Y, X\rangle_I = B(R(X, Y)Y, X) = -B([[X, Y], Y], X) = B([X, Y], [X, Y]) \le 0,$$

where the last step uses ad-invariance of B and the fact that $[X, Y] \in \mathfrak{so}(d)$, on which B is negative definite. Since the denominator of the sectional curvature is strictly positive for linearly independent X, Y, every sectional curvature is ≤ 0. By homogeneity (congruence isometries), the same sign holds at every $A \in \mathrm{SPD}^1(d)$. □

H. Hyperparameter sensitivity

Setup. We fix the GLSEA Lake Michigan setting (mesh, split, and all non-swept hyperparameters) and vary: (i) the ellipticity cap $s_{\max}$ for the unimodular metric field; (ii) a joint multiplier λ on the metric regularisers $(\texttt{reg_s_l2}, \texttt{reg_s_grad})$; and (iii) $(\alpha, \kappa_{\text{mult}})$ where $\kappa = \kappa_{\text{mult}}/L_{\text{char}}$ using the characteristic length reported by our tuning rule (see `tuning.json` in each run directory). Metrics are prediction RMSE and the geometry diagnostic Gap/AUC (Section 5.2). To keep the sweep lightweight, we use a single seed (seed 0), subsample to 5k train / 2k test points, reduce training to 30 steps, and use fewer probes for RMSE and DGC estimation; this preserves the same evaluation pipeline while keeping runtime small. In this setting, metrics are nearly insensitive to $\lambda \in \{0.5, 1, 2\}$. We generate Figure 2 via `scripts/hparam_sensitivity_glsea.py`.

Practical tuning recipe (reproducible).

1. **Pick κ from domain scale, then sweep κ_{mult}.** Use $\kappa = \kappa_{\text{mult}}/L_{\text{char}}$ and sweep a small grid (e.g., $\kappa_{\text{mult}} \in \{20, 50\}$). Choose larger κ_{mult} if you prioritise boundary-aware geometry (higher Gap/AUC), or smaller κ_{mult} if RMSE dominates.
2. **Choose an integer α first.** Start with $\alpha = 2$ (cheap and stable; avoids rational blocks when $J = 0$). Consider $\alpha = 3$ if you need a smoother prior; re-sweep $(\alpha, \kappa_{\text{mult}})$ jointly since their effects interact.
3. **Set $s_{\max}$ by stability.** Increase $s_{\max}$ to allow stronger anisotropy until training becomes unstable or linear solves become poorly conditioned; if this happens, reduce $s_{\max}$ and/or increase λ.
4. **Set regularisation on a log scale.** Multiply $(\texttt{reg_s_l2}, \texttt{reg_s_grad})$ by $\lambda \in \{0.5, 1, 2\}$; pick the smallest λ that prevents noisy, high-frequency artefacts in $A_\theta(x)$ while maintaining RMSE.

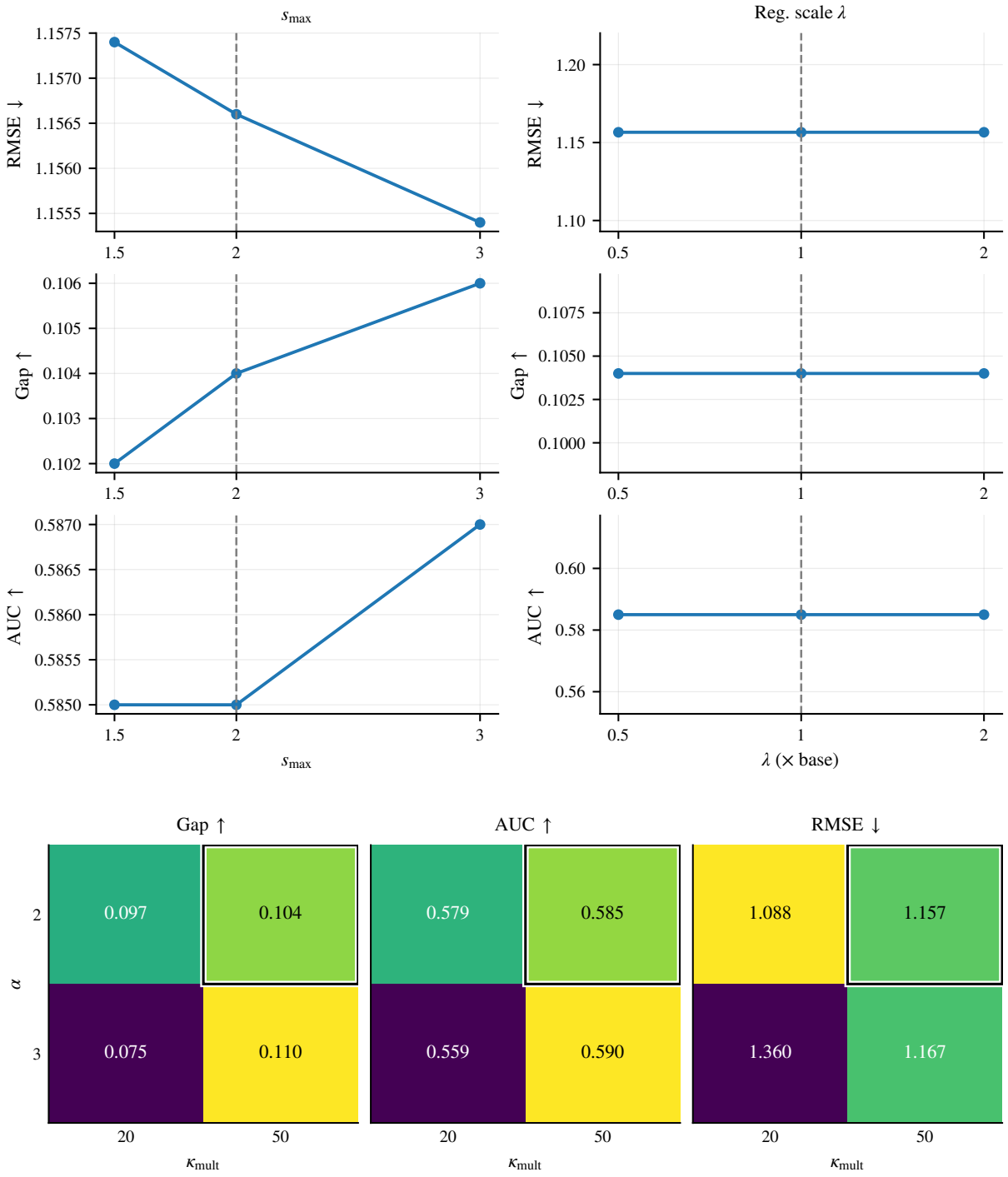

Figure 2. GLSEA (Lake Michigan): lightweight sensitivity sweep for RDM-Geo. Top: one-dimensional sweeps for $s_{\max}$ and for the regulariser multiplier λ (dashed line indicates the base setting). Bottom: a 2×2 grid over $(\alpha, \kappa_{\text{mult}})$ (outlined cell indicates the base setting). We report RMSE and the geometry diagnostic Gap/AUC.

I. Experiment details

I.1. GLSEA preprocessing and domain extraction

We access the NOAA GLSEA daily lake-surface temperature product via ERDDAP as `GLSEA_ACSPO_GCS`. From a subsampled grid we retain water pixels and convert longitude/latitude to local planar coordinates (metres). The domain Ω_{water} is extracted by applying marching squares (level 0.5) to the binary water mask, simplifying the resulting polygon loops, and meshing the interior for FEM. We sample a fraction of water pixels as training points and hold out the remainder for evaluation.

I.2. Protocol and fairness

For each experimental setting we fix preprocessing, mesh, train/test split, and method hyperparameters in a configuration file before running the reported 3-seed experiment (mean$\pm$std over seeds). Training uses only the designated training set. Evaluation-time numerical settings (e.g., probe counts or solver tolerances) are treated as numerical-accuracy knobs and are not tuned per seed.

All experiments are run on a single workstation (AMD Ryzen 9 5900X CPU; NVIDIA RTX 3090 GPU). RDM's sparse FEM/SPDE pipeline (assembly and PETSc Krylov solves) runs primarily on CPU, while the neural metric/density fields are trained on GPU in single precision (TF32 enabled on Ampere GPUs). The reported results do not rely on datacenter-grade FP64 GPUs; the dominant cost is sparse CPU linear algebra.